Infections & Inflammation

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · **P2** = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 26	Infections of the Urinary Tract
Ch 27	Inflammatory and Pain Conditions of the Lower Genitourinary Tract: Prostatitis, Orchitis, Epididymitis and Scrotal Pain
Ch 28	Interstitial Cystitis/Bladder Pain Syndrome and Related Disorders
Ch 29	Sexually Transmitted Diseases
Ch 30	Cutaneous Diseases of the External Genitalia
Ch 31	Tuberculosis and Parasitic Infections of the Genitourinary Tract
★	Putting it together: infection decisions

26 Infections of the Urinary Tract

1 · DEFINITIONS AND CLASSIFICATION

The label decides treatment length and whether to investigate.

- **Urinary tract infection (UTI)** is the urothelial inflammatory response to bacterial invasion, usually with bacteriuria and pyuria. Urine is **not sterile**.
- **Bacteriuria** and **pyuria** are not the same as UTI; bacteriuria without pyuria is colonisation. **Sterile pyuria**: look for tuberculosis (TB) see Ch 31, stone, urothelial injury and interstitial nephritis.
- **Cystitis** causes dysuria, frequency and urgency; mimics include urethritis, vaginitis, interstitial cystitis see Ch 28 and bladder carcinoma.
- **Acute pyelonephritis** is a clinical diagnosis: chills, fever and flank pain with bacteriuria and pyuria. Flank symptoms may be absent in spinal cord injury (SCI).
- **Chronic pyelonephritis** shows a focal coarse cortical scar over a **distorted calyx**.

	Uncomplicated UTI	Complicated UTI
Host	Healthy, non-pregnant woman	Compromised host and/or virulent or resistant organism
Tract	Structurally and functionally normal	Structural or functional abnormality, catheter
Organism	Eradicated by short oral course	Often multidrug-resistant (MDR), so culture is mandatory

- **Complicating factors** are an anatomical or functional abnormality (obstruction, neurogenic bladder [NGB], bladder outlet obstruction [BOO]), **male sex**, pregnancy, diabetes mellitus (DM), immunosuppression, catheter, hospital acquisition and symptoms >7 days. MCQ Male sex alone makes a UTI complicated.

Term	Definition	Discriminator
Isolated	First, or one remote UTI	—
Unresolved	Therapy failed; same organism	Never cleared

Term	Definition	Discriminator
Reinfection	New inoculation (~ **70%** of recurrences, often different organism)	Cleared, then new event
Bacterial persistence (relapse)	Same organism from a focus (stone, prostate)	Focus remains; recurs quickly

- **Prophylaxis** prevents reinfection of a sterilised tract; **suppression** prevents growth from an ineradicable focus.
- **Catheter-associated UTI (CAUTI)** is the commonest cause of secondary bacteraemia and ~**80%** of nosocomial UTIs.
- Uncomplicated recurrent UTI does not cause scarring, hypertension or renal failure.

2 · PATHOGENESIS

Route	Frequency	Features
Ascending	Commonest	Bowel flora colonise the introitus and climb; reflux not required
Haematogenous	Uncommon	S. aureus, Candida; enhanced by obstruction
Lymphatic	Rare	Severe bowel infection, retroperitoneal abscess

Organisms

- **E. coli** causes ~**85%** of community and ~50% of hospital UTIs, but less than half of complicated UTI and urosepsis.
- **S. saprophyticus** causes ~10% of cystitis in young women; complicated UTIs involve Klebsiella, Pseudomonas and E. faecalis.
- **ST131 (O25b:H4)** is an MDR E. coli clone carrying extended-spectrum β-lactamase (ESBL), with fluoroquinolone resistance as its hallmark. MCQ Enterococcus is a Gram-positive coccus, not an Enterobacteriaceae.

Virulence — fimbriae

	Type 1 pili (mannose-sensitive)	P pili (mannose-resistant)
Gene / adhesin	fim; **FimH** tip adhesin	pap operon on a pathogenicity island
Receptor	Mannosylated **uroplakins Ia/Ib**	Glycolipid (P blood-group) receptors in kidney
Tropism	Bladder, causing cystitis	Kidney, causing **pyelonephritis** (~ **90%** of pyelonephritis strains)

- Adherence is the essential first step. In **phase variation** bacteria switch between piliated and non-piliated states.

Intracellular cycle (why cystitis recurs)

ALGORITHM

1. FimH binds uroplakin and the bacterium invades the umbrella cell (15 min–1 h).
2. It multiplies into an **intracellular bacterial community (IBC)**, which behaves like a biofilm (4–16 h).
3. Bacteria filament, escape and reinvade neighbouring cells (16–24 h).
4. Infected cells exfoliate, exposing deeper layers.
5. **Quiescent intracellular reservoirs (QIRs)** form (from ~2 weeks).

- QIRs are silent between attacks; urothelial disruption reactivates them, giving same-strain recurrence without a urological focus.

Host defences and susceptibility

- Lactobacilli are the introital barrier, disturbed by antimicrobials, spermicides and oestrogen loss.
- Bladder emptying is the key defence; **obstruction** makes infection life-threatening. In adults, vesicoureteral reflux (VUR) alone is harmless without high pressure or stasis.
- **DM** brings more abscess, emphysematous infection and papillary necrosis.

Renal papillary necrosis

- The main causes are **DM, obstruction with pyelonephritis, analgesic abuse and sickle cell disease**. A sloughed papilla obstructs the ureter; treat with antibiotics and early relief of obstruction.

3 · EVALUATION AND DIAGNOSIS

A fistula presents with pneumaturia or faecaluria, and an abscess with indolent fever and a flank mass.

	Cystitis	Vaginitis	Urethritis
Onset	Acute	Subacute	Subacute
Frequency/urgency	Prominent	Absent; no haematuria or suprapubic pain	Less marked; no fever
Clues / causes	E. coli	Discharge, odour, new partner; HSV, gonorrhoea, Chlamydia, Trichomonas, yeast, bacterial vaginosis	Discharge; initial-stream pyuria in men; N. gonorrhoeae, Chlamydia, HSV see Ch 29

- Dysuria with frequency and no vaginal discharge gives a ~**90%** probability of UTI; dysuria is the best single symptom.
- **Rule**: painless gross haematuria, or microhaematuria persisting after the infection clears, needs full upper and lower tract evaluation.
- Midstream urine is the default; a catheter sample is more accurate. With an indwelling catheter, replace it and sample the new one.

Urinalysis and culture

- **Leucocyte esterase** indicates pyuria. **Nitrite** is made by Enterobacteriaceae and is negative with Gram-positives and Pseudomonas; it is the best rule-in test. Dipsticks are better at **ruling out**.
- On microscopy, pyuria is ≥**5–10 WBC/high-power field (HPF)**, and squamous cells mean contamination. A positive culture without pyuria should be repeated on a catheter sample.

Threshold	Use
≥10^5 colony-forming units (CFU)/mL	Kass (originally pyelonephritis); ASB definition
≥**10^2 CFU/mL** of a known pathogen	Symptomatic women with cystitis; catheter specimens

- **Stone culture** predicts post-procedure sepsis better than bladder urine.

Imaging

- Imaging asks only whether there is obstruction, a collection to drain or a focus of persistence; most UTIs need none.
- **Indications in pyelonephritis**: possible obstruction, risk of papillary necrosis, prior reconstruction, **no response after 72 h**, DM, polycystic kidneys on dialysis, NGB and unusual organisms. Also image most men, compromised hosts and rapid recurrences.

Question	Best test
Stone	Non-contrast computed tomography (CT)
Abscess, acute focal bacterial nephritis (AFBN), drainage planning	Contrast CT
Hydronephrosis, pyonephrosis, post-void residual (PVR)	Ultrasound (US)

Question	Best test
Extrarenal extension	Magnetic resonance imaging (MRI)
Urethral diverticulum	Pelvic MRI with contrast
VUR; non-refluxing stump	Voiding cystourethrogram (VCUG)
Split function, scarring	Dimercaptosuccinic acid (DMSA) scan

Correctable causes of bacterial persistence (Box 26.6) include infection stones, chronic bacterial prostatitis, an infected atrophic kidney, ectopic ureter, foreign body, urethral diverticulum, an infected ureteric stump and fistula. Antibiotics cannot sterilise these, so **cure comes only by removal or drainage.**

4 · ANTIMICROBIAL PRINCIPLES

- In cystitis the **urinary** drug level matters; in parenchymal infection or bacteraemia, **serum and tissue** levels matter. Renal insufficiency and obstruction lower urine levels.
- **Collateral damage** means resistance, MDR colonisation, C. difficile and candidiasis, caused mainly by broad cephalosporins and quinolones.

P1 *Antimicrobial resistance; ESBL producing enterobacterial UTI — intrinsic / chromosomal / plasmid*

resistance; ESBL, AmpC, KPC and their treatment options

Not in the paper answer:

- Prior quinolone use is the strongest determinant of quinolone resistance, and plasmid resistance spreads among bowel flora.
- **Stewardship** means narrow spectrum, the shortest course, culture before treating recurrent or complicated UTI, and local antibiograms.

Agents

- Fosfomycin acts on the cell wall at a different step from β-lactams, so cross-resistance is rare. Enterococci scavenge folate, so TMP-SMX fails even when "sensitive".
- **Anti-pseudomonal** agents are antipseudomonal penicillins, **ceftazidime** (not ceftriaxone), aztreonam, aminoglycosides and fluoroquinolones.
- **Enterococcal** cover comes from aminopenicillins, piperacillin, nitrofurantoin and fosfomycin, and **never from cephalosporins**.
- Ertapenem misses Pseudomonas and Enterococcus.

Agent	Key points
Nitrofurantoin	Acts in urine only, so it is useless for pyelonephritis, prostatitis and bacteraemia. Toxicity includes pulmonary fibrosis and neuropathy. Beers 2015: avoid if creatinine clearance (CrCl) **< 30 mL/min**; no long-term suppression in the elderly.
TMP-SMX	Good tissue levels; misses Enterococcus and Pseudomonas; potentiates warfarin.
Fosfomycin	3 g single dose; active against ESBL producers and vancomycin-resistant enterococci, so it is an oral option for MDR cystitis.
Fluoroquinolones	Broad, including Pseudomonas. Adjust at CrCl <30. FDA warnings include tendon rupture and aortic dissection; **2016: reserve for when there is no alternative in uncomplicated UTI**. Avoid in children and pregnancy.
Cephalosporins	Poor against enterococci; use in cystitis only if first-line agents cannot be used.
Carbapenems	Stable to ESBL and AmpC; penicillin cross-sensitivity <1%.
Aminoglycosides	Once daily 5–7 mg/kg; a nomogram level 6–14 h after the dose sets the interval at 24, 36 or 48 h. Monitor oto- and nephrotoxicity.
Aztreonam	Gram-negative spectrum without nephrotoxicity; useful in penicillin allergy and renal impairment.

Agent	Key points
Pivmecillinam	~2% E. coli resistance; Nordic first-line; not in USA.

KEY POINT

Narrow, urine-confined, flora-sparing agents (nitrofurantoin, fosfomycin, pivmecillinam) lead in cystitis; broad tissue-penetrating agents are kept for invasive infection. Resistance is made in the bowel, so breadth, duration and repetition matter more than MIC.

C. difficile infection (CDI)

- The usual culprits are cephalosporins, clindamycin, carbapenems, quinolones and TMP-SMX. Diagnosis needs ≥3 unformed stools in 24 h and toxigenic C. difficile or its toxin.

Episode (IDSA/SHEA 2021)	Treatment
Initial	**Fidaxomicin** 200 mg twice daily × 10 d; alternative oral vancomycin 125 mg four times daily × 10 d; metronidazole only if non-severe and others unavailable
First recurrence	Fidaxomicin or tapered vancomycin, with or without **bezlotoxumab** (anti-toxin B antibody)
≥2 recurrences	As above, or **faecal microbiota transplantation**
Fulminant	Oral/rectal vancomycin 500 mg four times daily and IV metronidazole

5 · ASYMPTOMATIC BACTERIURIA

Treating ASB usually harms; know the few groups who need it.

Group (Infectious Diseases Society of America [IDSA])	Specimen	Threshold
Women	**Two** consecutive voided, same strain	≥10^5 CFU/mL
Men	One voided	≥10^5 CFU/mL
Either sex	One catheterised	≥**10^2** CFU/mL

- Prevalence is 1–5% in premenopausal women and **100%** with a long-term catheter. Untreated ASB is harmless; treating it brings more symptomatic UTI and MDR.

Do NOT screen or treat (IDSA 2019)	Screen and treat
Premenopausal non-pregnant women; DM; elderly (community or institution); SCI; indwelling catheters; ASB with pyuria; children; renal transplant >1 month; other solid-organ transplant; **non-urological surgery (including arthroplasty and cardiac)**. No consensus: neutropenia.	**Pregnancy**; **urological procedures with anticipated mucosal bleeding** (transurethral resection [TUR], ureteroscopy, percutaneous nephrolithotomy [PCNL])

KEY POINT

ASB is treated only where bacteria are about to reach blood or tissue: pregnancy (hydroureter leading to pyelonephritis) and mucosa-breaching urological surgery. Elsewhere the resident strain is harmless or protective.

6 · CYSTITIS

Acute uncomplicated cystitis in women

- Diagnose on acute dysuria with a change in voiding. Fever, flank pain or no oral route should be treated as pyelonephritis.

Tier (IDSA/ESCMID)	Regimen	Caveat
First-line	**Nitrofurantoin** 100 mg twice daily × **5 d**	Not if pyelonephritis suspected
First-line	**TMP-SMX** 160/800 mg twice daily × **3 d**	Avoid if local resistance > **20%** or used in past 3 months
First-line	**Fosfomycin** 3 g single dose	Slightly lower efficacy
First-line	**Pivmecillinam** 400 mg twice daily × 3–7 d	Not in North America
Second-line	Fluoroquinolone × 3 d	Ecology and toxicity; last resort (FDA 2016)
Second-line	β-lactam (amoxicillin–clavulanate, cefpodoxime, cefadroxil) × 3–7 d	Lower efficacy; never amoxicillin alone

A **3-day** course beats a single dose and equals 7 days (7 if complicating factors). No post-treatment culture if asymptomatic.

Cystitis in men

- Male cystitis is complicated and about half have an abnormality, so evaluate and consider prostatitis see Ch 27.
- In afebrile men, **7 days is as good as 14 days** (Drekonja). The EAU suggests TMP-SMX 160/800 twice daily × 7 d, or a quinolone guided by local susceptibility.

Complicated UTI

Category (Canadian Urological Association [CUA])	Examples
Anatomical	Cystocele, fistula, diverticulum
Iatrogenic	Nosocomial, surgery, catheter
Obstruction	BOO, pelviureteric junction (PUJ) obstruction, ureteric stricture
Voiding dysfunction	Neurological disease, VUR, incontinence, high PVR
Other	Pregnancy, stones, DM, immunosuppression

Always culture and correct the abnormality. Treat **10–14 days** by culture, IV to oral once afebrile.

Emphysematous cystitis

- Bladder-wall gas, typically in an elderly diabetic woman; fever occurs in only 30–50%. **CT confirms** it.
- ~**90%** settle with broad Gram-negative cover, catheter drainage and glucose control.

Unresolved bacteriuria

Cause (descending frequency)	On-treatment culture	Action
1. Resistant from outset (recent antibiotics)	Resistant	Change drug; no further work-up
2. Emergent resistance (underdosing)	Same species, now resistant	
3. Unsuspected second pathogen	New species present from start	
4. Rapid reinfection, new resistant species	New species; suspect enterovesical fistula	
5–7. Azotaemia; papillary necrosis; giant staghorn	Same species, still **sensitive**	Renal function and imaging
8. Non-compliance, Munchausen	—	—

7 · RECURRENT UTI

Most recurrences are reinfections, so the work is prevention.

- Recurrent UTI means **2 in 6 months or ≥3 in 12 months**, each culture-documented. Reinfection is a different organism, or the same one >2 weeks after treatment.
- Risk factors include intercourse, spermicide, menopause, incontinence, cystocele and raised PVR.
- Evaluate with history, pelvic examination, **culture of each episode** and PVR. **Imaging and cystoscopy are not routine** (AUA, EAU, CUA).
- **Indications for imaging with or without cystoscopy (CUA)**: prior urinary surgery or trauma; stones; gross or persisting microhaematuria; obstructive symptoms or high PVR; urea-splitters; prior pelvic malignancy; persistence despite sensitivity-based therapy; DM or immunocompromise; pneumaturia, faecaluria or anaerobes; and repeated pyelonephritis. MCQ Recurrence frequency alone is **not** an indication.
- **Struvite** stones form when urease (Proteus mirabilis) produces magnesium ammonium phosphate, with bacteria inside the stone. The AUA advises complete removal in fit patients; unfit patients get acetohydroxamic acid and suppression.

Prevention (women, AUA 2019)

ALGORITHM

1. Confirm recurrent UTI with symptoms and culture.
2. If the same organism returns at short intervals, then evaluate for persistence (upper tract imaging and cystoscopy) and remove the nidus.
3. If it is reinfection, then start behavioural measures: ≥1.5 L water a day, frequent and postcoital voiding, stop spermicide.
4. Add cranberry ≥36 mg proanthocyanidin (PAC).
5. If UTIs follow intercourse, then give a **postcoital antibiotic**. If postmenopausal, then give **vaginal oestrogen**. If neither, then consider **methenamine hippurate** (normal renal function).
6. Use continuous antibiotic prophylaxis last.

Measure	Mechanism	Evidence / use
Water ≥1.5 L/day	Washout	RCT in premenopausal women: 1.7 vs 3.2 episodes/yr
Vaginal oestrogen	Restores lactobacilli, lowers pH	Strongest non-antibiotic evidence; ~90% reduction (Raz–Stamm). **Oral oestrogen is ineffective.**
Cranberry ≥36 mg PAC (twice daily)	Type A PAC blocks **P-fimbrial** adhesion	Cochrane 2023 favourable; safe with warfarin
Methenamine hippurate 1 g twice daily	**Formaldehyde in acid urine (pH <6)**	Non-inferior to daily prophylaxis; fails in NGB. Contraindicated if CrCl <50.
D-mannose	Inhibits **FimH**	Cochrane 2022 inconclusive
Immunostimulants (OM-89, Urovac, Uromune MV140)	Bacterial lysates	Some benefit; small biased trials
Probiotics, oral oestrogen, vitamin C	—	Not effective

Antibiotic strategy	Evidence	Position
Continuous low dose 6–12 months (nitrofurantoin, TMP, TMP-SMX, cephalexin)	Works only while taken; MDR	AUA "may prescribe"; EAU: last resort
Postcoital single dose	Effective; preferred for coital UTI	Acceptable

Antibiotic strategy	Evidence	Position
Self-start	Old data; overuse	Prefer analgesic, sample and observe, then culture-directed treatment

KEY POINT

Recurrent UTI in women is reinfection in a receptive host, not a hidden focus. Investigate only for features of persistence or complication; prevention targets flora, pH, adhesion and washout.

8 · KIDNEY INFECTIONS

Acute pyelonephritis

- **White cell casts** prove a renal origin. Take blood cultures if toxic, admitted or pregnant; the usual cause is P-fimbriated E. coli.

Setting (IDSA)	Regimen	Days
Outpatient, no vomiting	Ciprofloxacin 500 mg twice daily if community quinolone resistance ≤**10%** ; levofloxacin 750 mg daily; TMP-SMX double-strength twice daily only if known sensitive. With or without one IV dose of ceftriaxone 1 g or an aminoglycoside.	Cipro 7; levo 5; TMP-SMX 14
Inpatient, severe/possible sepsis	Ampicillin and gentamicin; levofloxacin; ceftriaxone 1 g daily; carbapenem (ESBL risk). Gram-positive cocci: ampicillin–sulbactam with or without an aminoglycoside.	10–14 total
Pregnancy	Ampicillin and gentamicin, aztreonam or ceftriaxone, then oral cephalexin. **No quinolone.**	10–14

- **Obstruction** needs urgent decompression by **stent or percutaneous nephrostomy**.

ALGORITHM

1. If there is no sepsis or vomiting, then send a culture, with or without US, and treat as an outpatient with oral drugs for 7–10 d.
2. If there is sepsis, then take blood and urine cultures, image, and admit for IV treatment.
3. If improved by **72 h**, then step down to oral treatment.
4. If not improved, then repeat cultures and get a **CT**, and drain any obstruction or abscess.

KEY POINT

Urine sterilises within hours but fever may lag for days. Fever beyond 72 h means obstruction or an undrained collection until CT proves otherwise. Antibiotics treat the infection; drainage treats the complication.

Acute focal bacterial nephritis (AFBN)

- AFBN is a lobar infiltrate between pyelonephritis and abscess. On contrast CT it is wedge-shaped with no wall or liquefaction, whereas an abscess is round with a liquid centre and a **ring sign**. Give IV antibiotics for ≥7 days, then 7 days oral.

Emphysematous pyelonephritis (EPN)

P1 P3 *EPN. Prognostic factors & conserv. mgt.; Prognostic indicators in emphysematous pyelonephritis*

— definition, CO_2 mechanism, CT choice, Wan patterns, Huang–Tseng risk factors, drainage-first management

Not in the paper answer:

- EPN is mostly seen in **diabetic women**; non-diabetics are usually obstructed, with less extensive disease.

CT pattern	Prognosis
Streaky/mottled gas, **no fluid**	50–60% mortality
Fluid with bubbly/loculated gas, or gas in collecting system	<20% mortality

Huang–Tseng class †	Extent of gas	Management †
1	Collecting system only (emphysematous pyelitis)	Antibiotics and percutaneous catheter drainage (PCD) or stent
2	Parenchyma, no extrarenal extension	Antibiotics and PCD
3A	Perinephric space	PCD if <2 risk factors; nephrectomy if ≥2 or PCD fails
3B	Pararenal space	
4	Bilateral or solitary kidney	PCD (nephron-sparing)

† The risk factors are thrombocytopenia, acute kidney injury (AKI), altered consciousness and shock.

> **KEY POINT**
> Fluid is a good sign and streaky gas a bad one: a contained collection can be drained; diffuse gas without fluid means parenchyma is necrosing faster than the host can wall it off.

Renal and perinephric abscess

P1 *Renal and perinephric abscess — haematogenous vs ascending, perinephric causes, insidious course, CT, empirical antibiotics, size-based drainage (≤3 / 3–5 / >5 cm), nephrectomy*

Not in the paper answer:

- A renal abscess **≤3 cm** is treated with antibiotics; one **>5 cm** needs percutaneous drainage, as does any size in an immunocompromised or non-responding patient.
- A perinephric abscess has a lower drainage threshold (>3 cm); only a third are diagnosed at presentation, and delay causes nearly all deaths.

Infected hydronephrosis and pyonephrosis

- **Pyonephrosis** is infected hydronephrosis with suppurative destruction. Patients are very ill, **bacteriuria may be absent** if the ureter is blocked, and US shows a fluid–debris level.
- Give antibiotics with **immediate drainage** (stent or nephrostomy); nephrectomy if non-functioning.

XGP and malakoplakia

P1 P2 *Xanthogranulomatous pyelonephritis; Malakoplakia — stone–obstruction–Proteus triad, bear-paw CT, RCC mimic, complete nephrectomy; Michaelis–Gutmann bodies, intracellular antibiotics + bethanechol/vitamin C*

Not in the paper answer:

- In xanthogranulomatous pyelonephritis (XGP), urine is sterile in about a third but tissue culture is positive.

	XGP	Malakoplakia
Cell	Foamy lipid-laden macrophages	Von Hansemann cells, Michaelis–Gutmann bodies
Stones/obstruction	Usually	Absent
Organism	Proteus, E. coli	E. coli
Laterality	Unilateral	Uni-/multifocal, may be bilateral
Treatment	Nephrectomy	Intracellular antibiotics with or without nephrectomy

KEY POINT

Across renal infections the axis is diffuse inflammation versus contained collection. Pyelonephritis and AFBN need antibiotics; pyonephrosis and large or obstructed abscesses need drainage; when the kidney itself is the infected tissue (XGP, diffuse EPN), the answer is nephrectomy.

9 · FOURNIER GANGRENE AND PERIURETHRAL ABSCESS

Skin hides the true spread; only early radical debridement saves lives.

- **Fournier gangrene** is necrotising fasciitis of the genitalia, perineum or perianal region. A source is found in ~**95%**, and the male:female ratio is ~10:1.
- Sources are **colorectal**, **urogenital** (stricture, instrumentation, catheter) and **dermal**. Risk factors are **DM**, obesity, alcoholism and immunosuppression.
- Synergistic organisms cause obliterative endarteritis, spreading along Buck's, dartos, Colles' and Scarpa's fascia.
- Pain out of proportion comes first, then crepitus and gangrene. **The skin underestimates fascial extent**, and CT must not delay surgery.

Score	Components	Use
Laboratory Risk Indicator for Necrotising Fasciitis (LRINEC)	CRP, WBC, haemoglobin, sodium, creatinine, glucose	≥**6** suspicious; ≥**8** strongly predictive
Fournier's Gangrene Severity Index (FGSI) †	Nine vital signs and blood values, each scored 0–4	>9 predicts mortality

ALGORITHM

1. On suspicion, resuscitate and start broad IV antibiotics (piperacillin–tazobactam; or a carbapenem or vancomycin with clindamycin or metronidazole).
2. Perform **immediate radical debridement** back to bleeding fascia and leave the wound open.
3. Take the patient back for a **re-look at 24–48 h**.
4. If the source is urethral, then add suprapubic diversion; if rectal, then a colostomy.
5. Consider hyperbaric oxygen as an adjunct, then reconstruct.

- **Orchidectomy is almost never needed**, because the testicular blood supply is independent of the scrotal skin. Mortality is ~20–40%.
- A **periurethral abscess** needs drainage, suprapubic diversion and antibiotics.

10 · UROSEPSIS †

Term (Sepsis-3)	Definition
Sepsis	Organ dysfunction from a dysregulated response to infection: rise in Sequential Organ Failure Assessment (SOFA) score ≥**2**
Quick SOFA (qSOFA)	≥2 of: respiratory rate ≥22/min, altered mentation, systolic BP ≤100 mmHg
Septic shock	Vasopressors for mean arterial pressure (MAP) ≥65 mmHg **and** lactate >2 mmol/L despite fluids

ALGORITHM

1. Hour-1 bundle: measure lactate and take blood and urine cultures.
2. Give a broad IV antibiotic within 1 h.
3. If hypotensive or lactate ≥4, then give 30 mL/kg crystalloid.
4. If still hypotensive, then start noradrenaline for MAP ≥65.
5. Achieve **source control**: decompress with a stent or nephrostomy; no definitive stone surgery now.
6. De-escalate antibiotics by culture.

† Not in the source notes. Manipulating infected stones releases endotoxin, so drain first and operate later.

11 · SPECIAL POPULATIONS

Pregnancy

P1 P3 *Bacteriuria in pregnancy; Baceriuria and use of antibiotic policy in pregnancy — prevalence unchanged but ascends, culture screen at 12–16 weeks, 4–7 day course, safe/avoid drug table, pyelonephritis and suppression*

Not in the paper answer:

- GFR rises by 30–50%, so creatinine falls to about **0.5 mg/dL**, and a "normal" 0.8–0.9 may mean impairment.
- Severe or MDR pyelonephritis needs a carbapenem.

Elderly

- **Do not screen for or treat ASB**; bacteriuria with confusion is not UTI. The **absence** of pyuria **excludes** bacteriuria.
- The **revised McGeer criteria** require **localising** signs and ≥10^5 CFU/mL of ≤2 species in non-catheterised residents; catheterised residents may qualify on **systemic** signs.
- Treat for 3–6 days if uncomplicated and 10–14 if febrile.

Catheter-associated UTI

P1 *Catheter Associated Urinary Tract Infections (CAUTI) — definition, 5–10% per catheter-day, extra-/intraluminal routes, biofilm, prevention bundle, change catheter, treat only symptomatic*

Not in the paper answer:

- Catheters in place >30 days carry 4–5 species, and silicone is no better than latex.

Spinal cord injury

- All UTIs in SCI are complicated, and UTI is the commonest cause of fever. Symptoms are atypical: fever, spasticity, leakage, autonomic dysreflexia † and cloudy urine.
- **Clean intermittent catheterisation (CIC)** has the fewest long-term complications and an indwelling urethral catheter the most.

Drainage (NIDRR)	Significant bacteriuria
Indwelling urethral or suprapubic catheter	Any detectable count
Intermittent catheterisation	≥10^2 CFU/mL
Catheter-free men (clean voided)	≥10^4 CFU/mL

- Do not treat ASB. Treat afebrile UTI for 5–7 days; febrile UTI needs admission and **14 days** parenteral-first treatment. Change the catheter.
- Prophylaxis does not work in NGB; for recurrent UTI, get imaging and urodynamics (UDS) and lower storage pressure.
- A catheter for >**10 years** raises the risk of bladder cancer, so these patients need surveillance.

Diabetes, renal transplant

- In DM, do not treat ASB, but keep a low threshold to image pyelonephritis.
- After transplant, do not screen for ASB beyond 1–2 months. Treat uncomplicated UTI for 5–10 days and pyelonephritis for 14–21 days. Stents raise UTI by about half, so remove them within **4 weeks**.

12 · FUNGAL UTI

P1 *Fungal UTI — Candida risk factors, asymptomatic candiduria (remove catheter; treat only neutropenia, VLBW infants, pre-procedure), fluconazole 2 weeks, amphotericin for resistant species, fungus ball drainage, IDSA vs AUA pre-procedure*

Not in the paper answer:

- AUA BPS 2019 advises antifungal prophylaxis for high-pressure endoscopy (TURBT, TURP, therapeutic ureteroscopy) and open entry into the tract.

13 · PERIOPERATIVE ANTIMICROBIAL PROPHYLAXIS

One well-timed dose suffices; more only adds resistance.

Principle (AUA BPS 2019)	Rule
Indication	Any procedure breaching a tissue barrier, weighing host defence and consequences
Timing	Within **60 min** of incision; vancomycin and quinolones within 120 min
Duration	**Single dose**; ≤**24 h** only for PCNL, prostheses, special cases
Agent	Local flora per antibiogram
Known infection	Sterilise urine before elective surgery (this is treatment); infected urine causes bacteraemia in ~half

- Host risk factors include age, smoking, steroids, immunodeficiency, hardware and infected stones.

Wound class	Examples	Antimicrobial prophylaxis (AP)
I clean	Scrotal/inguinal, retroperitoneal lymph node dissection (RPLND), nephrectomy without entering tract	Not needed (except groin/perineal incision, prosthesis)
II clean-contaminated	Any endoscopy; opening the urinary tract	Single dose
III contaminated	Infected stone PCNL, **transrectal biopsy**, bowel segments	Strongly recommended; anaerobic cover
IV dirty	Abscess, open trauma	Treatment, tailor to cultures

Procedure	AP?	Agent	Duration
Diagnostic cystoscopy / UDS (no risk factors)	No (EAU strongly against)	—	—
Cystoscopy with biopsy/fulguration	Host factors	TMP-SMX, amoxicillin–clavulanate, cephalosporin	Single
TURP / TURBT, laser prostate	All (TURBT: EAU only if high risk)	Cefazolin, TMP-SMX	Single (TURP short course with catheter acceptable)
Transrectal prostate biopsy	All	Fluoroquinolone or cephalosporin, with or without aminoglycoside; targeted by rectal swab	Single (≥1 day)
Ureteroscopy	All (diagnostic: uncertain)	TMP-SMX, 1st/2nd-gen cephalosporin	Single

Procedure	AP?	Agent	Duration
PCNL	All	1st/2nd-gen cephalosporin; aminoglycoside and metronidazole	≤24 h
Shock-wave lithotripsy (SWL)	Only with risk factors (EAU: no)	TMP-SMX, cephalosporin	Single
Open/robotic entering tract	All	Cefazolin	Single
Colon segment	All	Cefazolin and metronidazole / cefoxitin / ertapenem, plus oral neomycin with erythromycin or metronidazole	Single
Artificial urinary sphincter (AUS), penile prosthesis, sacral neuromodulation	All	Aminoglycoside and cephalosporin or vancomycin	≤24 h
Vaginal surgery incl. slings	All	Cefazolin/cefoxitin	Single
Penile (circumcision), clean scrotal	Likely none (scrotal: high-risk only)	Cefazolin	Single

- **Transrectal ultrasound (TRUS) biopsy** causes sepsis in 0.3–3.1%, mainly from quinolone-resistant E. coli. Use augmented or **targeted** prophylaxis; the AUA advises alternatives if local resistance is >20%.
- **UDS (SUFU 2017)**: give AP for neurogenic lower urinary tract dysfunction, raised PVR, ASB, immunosuppression, age >70, catheter or recent joint implant.
- **Catheter or stent removal**: give a single dose only if risk factors are present.
- **Endocarditis** (AHA): no AP solely to prevent it; with a high-risk valve and colonised urine, sterilise the urine first.
- **Orthopaedic hardware**: only a joint replaced <2 years ago or a high-risk host needs AP for high-bacteraemia procedures.

> **KEY POINT**
>
> Prophylaxis is justified by wound contamination × host vulnerability. Where neither is raised (diagnostic cystoscopy, sterile-urine SWL, index UDS), ecological cost outweighs benefit; otherwise give one well-timed, flora-matched dose. Anything longer is treatment.

14 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Raz & Stamm 1993	RCT, intravaginal oestriol vs placebo, postmenopausal recurrent UTI	0.5 vs 5.9 UTIs/patient-year; lactobacilli returned, pH fell	Vaginal oestrogen first-line after menopause
Hooton 2018	RCT, +1.5 L water/day vs none, premenopausal	Cystitis 1.7 vs 3.2/year	Hydration halves recurrences
Cai (ASB in recurrent UTI)	RCT, treat vs observe ASB in young women	Treatment increased recurrence and resistant E. coli	Do not treat ASB; E. faecalis may protect
ALTAR (methenamine)	Methenamine vs daily antibiotic prophylaxis, 12 months	Non-inferior	Non-antibiotic alternative
Drekonja 2021	RCT, afebrile male UTI, 7 vs 14 days ciprofloxacin/TMP-SMX	7 days non-inferior	7 days for afebrile men

27 Inflammatory and Pain Conditions of the Lower Genitourinary Tract: Prostatitis, Orchitis, Epididymitis and Scrotal Pain

1 · PROSTATITIS: DEFINITIONS AND CLASSIFICATION

Only categories I and II are infections; most men have category III.

- The Drach (1978) scheme, built on Meares–Stamey cultures, was replaced by the MCQ National Institutes of Health (**NIH, 1999**) classification.

NIH category	Entity	Defining feature
I	Acute bacterial prostatitis (ABP)	Acute prostatic urinary tract infection (UTI), often febrile
II	Chronic bacterial prostatitis (CBP)	Recurrent UTI, **same organism**; well between episodes
IIIA	Chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS), inflammatory	Leucocytes in expressed prostatic secretion (EPS), post-massage urine (VB3) or semen
IIIB	CP/CPPS, non-inflammatory	No leucocytes
IV	Asymptomatic inflammatory	Incidental on biopsy or semen analysis

- I and II are **5–10%**; III is **90–95%**. **IIIA vs IIIB has no clinical significance**, and the prostate may not be the pain source.
- **Category III (NIH CPCRN)** needs pelvic pain for **≥3 of the previous 6 months**, excluding uropathogens, urogenital cancer, prior radiation, stricture and neurological bladder disease.
- Histological inflammation is near-universal (**>70%** at autopsy), so it cannot define disease; **Nickel grading** records its location, extent and grade.
- **Granulomatous prostatitis** classically follows **intravesical bacillus Calmette–Guérin (BCG)**, giving a **transient benign prostate-specific antigen (PSA) rise**.

2 · CATEGORY I: ACUTE BACTERIAL PROSTATITIS

- Spread is mainly **ascending urethral**. Risks include catheters, instrumentation or biopsy, obstruction and insertive anal intercourse.
- **E. coli causes 65–80%**; Pseudomonas is commoner after instrumentation. After biopsy, organisms are often fluoroquinolone (FQ)-resistant or extended-spectrum β-lactamase (ESBL) producers.

> **KEY POINT**
>
> **Why young healthy men get ABP:** the organism, not the host — virulent uropathogenic E. coli that forms **biofilm** more often than cystitis strains; biofilm blocks antibiotics, so prostatic infection needs longer, well-penetrating therapy.

- Symptoms are frequency, dysuria, perineal pain, fever and rigors, with possible retention or sepsis. In febrile male UTI, **>90%** show raised PSA or prostate volume.

Evaluation

- First ask whether it was **spontaneous or after manipulation or biopsy**, as this decides empirical therapy.
- The prostate is tender and boggy; examine **gently and never massage**. Send midstream and **blood cultures**, and measure residual urine by **ultrasound (US)**.
- **No PSA** acutely; if raised, follow it to normal (up to **6 months**). Use transrectal ultrasound (TRUS) or CT only for suspected abscess, and **never biopsy**.

Treatment

- **Outpatient care only if** the man is not systemically ill, tolerates oral drugs and is not in retention.
- Systemic illness (**EAU**): amoxicillin or a 2nd-generation cephalosporin with an aminoglycoside, or a 3rd-generation cephalosporin. Ciprofloxacin only if local resistance **<10%** and fully oral outpatient care, or β-lactam anaphylaxis.
- **Febrile prostatitis after biopsy** (AUA White Paper): **no FQ or trimethoprim–sulfamethoxazole (TMP-SMX)**; use a **carbapenem**, amikacin or 2nd/3rd-generation cephalosporin.
- **Duration**: traditionally 4 weeks, but **2 weeks** of ciprofloxacin is equivalent; TB prostatitis needs ≥**6 months**. NSAIDs and α-blockers are useful adjuncts.
- **Retention**: brief urethral catheterisation is acceptable, but use a **suprapubic catheter for long-term** drainage.

Prostatic abscess

- Suspect it with high fever, immunosuppression or **no response after 48 h**. It is hypoechoic on TRUS, and **CT** shows spread.

ALGORITHM

1. If the abscess is <1–2 cm, then treat with antibiotics.
2. If it progresses and lies peripherally, then perform US-guided percutaneous drainage (recurrence 15–33%).
3. If it is large or drainage fails, then perform **transurethral unroofing**, using intraoperative US to locate it.
4. If it extends outside the prostate, then drain it openly (rare).

3 · CATEGORY II: CHRONIC BACTERIAL PROSTATITIS

Recurrent same-organism UTI in men needs a long penetrating course.

- The prostate harbours the same organism and the man is **well between episodes**, whereas CP/CPPS pain is continuous.
- Organisms are Gram-negatives and E. faecalis; biofilm explains the poor response. The role of **Chlamydia** is debated.

Localisation test	Specimens	Interpretation
Meares–Stamey 4-glass (previous)	VB1 urethra; VB2 bladder (midstream); EPS; VB3 post-massage	Prostatic if EPS/VB3 counts exceed VB1/VB2
2-glass pre/post-massage (current)	Midstream pre-massage and first 10 mL post-massage	Predicts 4-glass in **>95%**

- There is no agreed cut-off, though some use a **10-fold rise**; **do not culture semen**.
- A residual volume **>180 mL** raises infection risk. Men **<45** with a first UTI have normal imaging, but check for **stricture**.

KEY POINT

Why fluoroquinolones? Prostatic entry is passive (lipid solubility, ionisation, protein binding, size). FQs are lipid-soluble **zwitterions** that concentrate in alkaline CBP fluid and cover Gram-negatives, Gram-positives and atypicals. Tetracyclines, macrolides and trimethoprim also penetrate.

Situation (EAU)	Regimen
First line	FQ (ciprofloxacin = levofloxacin) **4–6 weeks**
FQ-resistant, TMP-SMX-sensitive	TMP-SMX **3 months** (cure 30–50% vs 60–80% with FQ)

Situation (EAU)	Regimen
C. trachomatis	Azithromycin 500 mg daily 3 weeks, or doxycycline 100 mg twice daily 10 days; azithromycin is better than ciprofloxacin
T. vaginalis	Metronidazole 500 mg three times daily 14 days

- A macrolide can be added because it **penetrates biofilm**. Refractory cases need suppression or TURP.

4 · CP/CPPS — AETIOLOGY

An insult (infection, stress, trauma) in a susceptible man starts **neurogenic inflammation**, which other factors maintain, so single treatments fail.

Domain	Key evidence
Infection	Bacterial counts are equal in patients and controls; in mice, pain persists after clearance
Inflammation	Biopsy inflammation does not correlate with symptoms; symptoms track **cytokines**
Neurological	**Central sensitisation**
Pelvic floor	Tenderness and **failure to relax**; tender in **81%**
Psychosocial	**Catastrophising and helplessness predict** pain
Endocrine	Blunted adrenocorticotropic hormone (ACTH) response

5 · CP/CPPS — CLINICAL FEATURES AND EVALUATION

P1 P4 *Chronic Prostatitis: diagnosis, classification and management; Chronic non bacterial prostatitis —*
pain sites, NIH-CPSI domains, DRE/trigger points, 2-glass test, cytology, cystoscopy, UPOINT-directed multimodal therapy

Not in the paper answer:

- The **NIH Chronic Prostatitis Symptom Index (NIH-CPSI)** † scores 0–43 (pain 0–21, urinary 0–10, quality of life 0–12), and a **≥6 points** fall is meaningful.
- **PSA is not indicated**, and a raised value must never be blamed on CP/CPPS.
- In men with bladder pain, about 42% meet interstitial cystitis criteria see Ch 28, and Hunner lesions respond to fulguration.

UPOINT phenotyping

Monotherapy fails because each man has a different mix of domains. UPOINT maps them for simultaneous treatment, and the domain count tracks severity.

Domain	Identify by	Therapy
Urinary	Lower urinary tract symptoms (LUTS), PVR	α-blocker; †antimuscarinic
Psychosocial	Depression, catastrophising	Psychology, stress reduction
Organ-specific	Prostate tenderness; bladder pain	†Quercetin/pollen; bladder therapy; 5α-reductase inhibitor (5-ARI) in BPH
Infection	Positive VB3 with negative VB2	Antibiotics
Neurologic/systemic	Pain beyond pelvis, irritable bowel syndrome (IBS), fibromyalgia	Tricyclics, pregabalin, acupuncture
Tenderness	Pelvic floor trigger points	Physiotherapy, myofascial release

6 · CP/CPPS — TREATMENT

KEY POINT

Governing fact: no monotherapy has a positive definitive trial, because each tested one agent in an unphenotyped population. Start conservative, then give **simultaneous, phenotype-directed multimodal therapy** with a multidisciplinary team.

Modality	Evidence / role
Antibiotics	Quinolone trials negative. EAU: quinolone or tetracycline ≥**6 weeks** only if **treatment-naive and <1 year**. **No repeated courses without a positive culture**.
α-blockers	Trials negative; use in newly diagnosed men with voiding symptoms. Best combination: **antibiotic and α-blocker**.
Anti-inflammatories	Celecoxib helps pain only; steroids negative.
5-ARI	Dutasteride (REDUCE): ≥6-point NIH-CPSI fall in 49% vs 37%; for older men with BPH.
Neuropathic	Pregabalin missed its endpoint; amitriptyline needs ≥50 mg.
Conservative	Diet, **aerobic exercise** and cognitive therapy.
Acupuncture	Strongest conservative evidence: sham-controlled RCT, 61.5% vs 38.3% responders at 32 weeks.
Pelvic floor physiotherapy	Myofascial trigger-point release beats global massage (57% vs 21%); refer if the pelvic floor is tender.
Neuromodulation	Tibial or sacral nerve stimulation if frequency/urgency.
Surgery	No TURP or prostatectomy. **Bladder neck incision** only for proven obstruction.

7 · ACUTE ORCHITIS

- **Bacterial** orchitis (age >35) is usually Gram-negative. **Mumps** is the commonest viral cause, following parotitis, with **atrophy in about half** but rare infertility. Autoimmune and vasculitic forms are treated with steroids.
- Ask about **pain duration**, because torsion is the key differential; get urine culture, STI tests and **scrotal US**. **Follow-up is mandatory** to avoid missing torsion or tumour.

8 · ACUTE EPIDIDYMITIS

Age and sexual history predict the organism.

- Epididymal pain and swelling for **<6 weeks**; infection is **ascending**.
- **In sexually active men <35, C. trachomatis and N. gonorrhoeae dominate**; **in men >35 and children, enteric organisms dominate**. Because >40% of men under 35 grow enterics, **screen all sexually active men for STI**.
- Send culture, first-void urine and STI NAAT; **Doppler US** excludes torsion.

Likely pathogen	Regimen
Chlamydia/gonorrhoea	Ceftriaxone **500 mg IM** (intramuscular) single dose (1 g if ≥150 kg) and doxycycline 100 mg twice daily **10 days**
STI and enteric (insertive anal sex)	Ceftriaxone 500 mg IM and levofloxacin 500 mg once daily 10 days
Enteric only	Levofloxacin 500 mg once daily 10 days

- Give rest, elevation and NSAIDs, and treat partners. With **no improvement at 72 h**, repeat US for abscess or infarction.

9 · CHRONIC ORCHITIS AND EPIDIDYMITIS

- **Granulomatous orchitis** has infective causes (TB, brucellosis, syphilis) and an **idiopathic** form usually found after orchidectomy for suspected tumour. **Malakoplakia** shows **Michaelis–Gutmann bodies**.
- **Chronic epididymitis** (≥**3 months**) has a **Nickel classification**: inflammatory (infective, TB, **amiodarone**, idiopathic), obstructive (congenital, post-infective, post-vasectomy) and epididymalgia (normal epididymis).
- **TB epididymitis** is the commonest genital site, presenting as painless swelling with a **sinus**. Give four drugs for ≥**6 months**, and operate **only on anti-TB drugs**. **A mass not shrinking after 3 weeks needs inguinal exploration**.

10 · CHRONIC SCROTAL CONTENT PAIN (CSCP)

A cord block predicts who benefits from denervation surgery.

CSCP is intrascrotal pain for ≥**3 months** that interferes with life; there are no evidence-based guidelines.

- Post-procedural causes are **inguinal hernia repair** (commonest) and vasectomy (~**1 in 1000**, the consent figure). Others include varicocele, intermittent torsion and referred pain from a ureteric stone.
- **Wallerian degeneration** affects the **"trifecta nerve complex"**: cremaster, perivasal sheath and posterior lipomatous tissue. Pelvic floor dysfunction is common.
- **First decide** whether the pain is isolated or part of CP/CPPS, and get a **duplex US**.
- A **spermatic cord block** uses **20 mL 0.25% bupivacaine** at the **pubic tubercle**. **>50% relief** predicts success of microsurgical denervation of the spermatic cord (MDSC).
- Medical options are NSAIDs and **neuropathic agents**, without chronic opioids. Botulinum toxin equalled placebo in an RCT.

ALGORITHM

1. If there is an identifiable lesion, then do directed surgery (hydrocelectomy, spermatocelectomy, orchidopexy, **microsurgical varicocelectomy**).
2. If the pain followed vasectomy, then offer **vasectomy reversal** (82–93%; wait ≥3 months).
3. If there is no lesion and the block is positive, then perform **MDSC**.
4. If the epididymis is focally affected, then perform epididymectomy.
5. Last resort: **inguinal orchidectomy**.

- **MDSC** is subinguinal and microscopic: divide the cord nerves, **preserve arteries and lymphatics** and ligate the internal spermatic veins. It cures 50–70%; consent for **testicular atrophy**. **Targeted trifecta MDSC** is equally effective and quicker.
- **Epididymectomy** succeeds in 50–75%. **Orchidectomy** is least reliable, and **inguinal beats scrotal** (75% vs 55%).

11 · LANDMARK TRIALS

Trial	Design	Result	Take-home
NIH CPCRN ciprofloxacin/tamsulosin	RCT, 6 weeks, NIH-CPSI ≥15	No benefit vs placebo	Monotherapy fails in unselected CP/CPPS
REDUCE (CP/CPPS analysis)	Dutasteride vs placebo, 4 years	≥6-point fall 49% vs 37%	5-ARI for older men with BPH
MAPP myofascial physiotherapy	Feasibility vs global massage	57% vs 21% responders	Physiotherapy for tender pelvic floor

Trial	Design	Result	Take-home
Acupuncture vs sham	Large RCT	61.5% vs 38.3% at 32 weeks	Evidence-based option
Botulinum toxin cord injection	Double-blind RCT	No better than local anaesthetic	Open-label series overestimate benefit

28 Interstitial Cystitis/Bladder Pain Syndrome and Related Disorders

1 · DEFINITION AND NOMENCLATURE

P2 P4 *Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management — AUA, ESSIC (types 1–3, A–C), NIDDK, ICD-11 definitions; IC vs BPS split; phenotypes*

Not in the paper answer:

- **IC/BPS** ranges from inflammatory disease with **Hunner lesions (HLs)** to pain with a normal bladder and many non-urological complaints.
- The **AUA** defines it as bladder-related pain, pressure or discomfort with lower urinary tract symptoms (LUTS) for **>6 weeks**, with no infection or other cause. † **ESSIC (2008)** requires **>6 months** and ≥1 urinary symptom; **type 3C** is classic HL disease.
- The **NIH/NIDDK (1987)** research criteria missed **~60%** when used clinically. The current **trend** is to call HL disease "IC" and non-HL disease "BPS".

2 · EPIDEMIOLOGY (HEADLINE)

- The RICE study found that **3.3–7.9 million US women** and 2.1–4.6 million men have IC/BPS-type symptoms.
- Female:male is **5:1** in clinics but **2–3:1** in populations, so men are under-diagnosed. **Ask men labelled CP/CPPS about bladder pain** see Ch 27.
- **HL prevalence** is **~20%**, rising with age.
- The course is chronic but **not progressive**, with flares, commonly after sexual activity.

> **KEY POINT**
>
> Uncontrolled series "work" but randomised controlled trials (RCTs) fail because of regression to the mean, placebo and extra attention. Expect improvement in the first 3–4 weeks of any treatment.

3 · AETIOLOGY AND PATHOPHYSIOLOGY

The cause is multifactorial, and each theory points to a treatment.

Theory	Mechanism / evidence	Treatment link
Glycosaminoglycan (GAG) layer / "leaky" urothelium	Urinary solutes such as potassium reach suburothelial nerves	Pentosan polysulfate (PPS), heparin, hyaluronic acid
Mast cells	Histamine release; marked in HL disease, but counts have **no diagnostic role**	Hydroxyzine, cimetidine, amitriptyline
Neurogenic	Neurogenic inflammation, pelvic cross-sensitisation and central sensitisation	Tricyclics, gabapentinoids, neuromodulation, CBT
Autoimmune	Excess of lupus, Sjögren syndrome and inflammatory bowel disease (IBD); lymphoplasmacytic infiltrate in HL	Ciclosporin A (CyA) for HL disease

Theory	Mechanism / evidence	Treatment link
Infection	No consistent organism; 18-week antibiotic RCT showed no benefit	No antibiotics without proven infection

KEY POINT

Central sensitisation explains why many BPS patients have no physical findings and do not respond to local bladder therapy — the mechanism is systemic, not vesical.

4 · PATHOLOGY: HUNNER LESION VS NON-HUNNER PHENOTYPE

- There is **no pathognomonic histology**. Biopsy only HLs or suspicious lesions, to exclude carcinoma in situ (CIS), tuberculous and ketamine cystitis.

Feature	HL disease ("IC")	Non-HL ("BPS")
Cystoscopy	Stellate erythema, radiating vessels, central scar; bleeds on filling ("waterfall"); rare at trigone	Normal, or glomerulations after distension
Histology	Ulceration, granulation, dense **lymphoplasmacytic** infiltrate, increased mast cells	Near-normal; sparse inflammation
Pain source	Bladder — end-organ disease	Often regional/widespread
Treatment	Fulguration / triamcinolone; CyA; dimethyl sulfoxide (DMSO) works better; cystectomy succeeds	Phenotype-directed; surgery often fails

5 · ASSOCIATED CONDITIONS

- **75–80%** have pain beyond the pelvis. Among overlapping pain conditions, **irritable bowel syndrome (IBS)** is commonest (>30%), followed by **fibromyalgia**, chronic fatigue and migraine.
- **High-tone pelvic floor dysfunction (HTPFD)** occurs in **~80%**. **Autoimmune** disease shows the largest relative excess; in lupus, think of **lupus cystitis**.

KEY POINT

Commonest is not the same as most over-represented: IBS is commonest; autoimmune disease shows the greatest relative excess. Widespread pain predicts failure of bladder-directed therapy.

6 · CLINICAL FEATURES

Pain linked to filling separates IC/BPS from overactive bladder.

- Suprapubic pain is **worse with filling and relieved by voiding**, driving day and night frequency.

Feature	Overactive bladder (OAB)	IC/BPS	HTPFD
Urge	Sudden; fear of leakage	Persistent, mounting, pain-driven	Persistent; pelvic pressure
Pain	Absent	Present	Present
Incontinence	Urgency incontinence	Not a feature	—
Voiding	—	Frequency, low volumes	Obstructive

7 · DIAGNOSIS — EXCLUSION PLUS LOCALISATION

Basic evaluation (AUA 2022)

- Ask about the pain–filling link, triggers, pelvic irradiation, **cyclophosphamide and ketamine**.
- Examine for vaginal and urethral pathology, do a **Q-tip test** and DRE, and **palpate the pelvic floor**.
- Test post-void residual (PVR), urinalysis and culture, with cytology if at risk. Score pain and urinary symptoms separately, for example with the O'Leary–Sant indices (trial endpoint ≥30% fall).

Confusable disorders (ESSIC-modified)

Group	Disorder	Excluded by
Malignancy	Bladder cancer/CIS; prostate, gynae, colon	Cystoscopy with or without biopsy; prostate-specific antigen (PSA); pelvic exam
Infection	Bacterial, atypical, **TB**, herpes; prostatitis	Culture; **sterile pyuria needs TB culture**
Iatrogenic/drug	Radiation, cyclophosphamide, tiaprofenic acid, **ketamine**	History; ketamine causes upper-tract change
Outflow	Bladder neck obstruction, benign prostatic obstruction, retention	PVR, uroflow, (video)urodynamics
Structural/other	Stones, diverticulum; endometriosis, vulvodynia, atrophy; IBS, IBD	Imaging, cystoscopy, exam, laparoscopy, colonoscopy
Neuromuscular	Pudendal entrapment; pelvic-floor myalgia	Nerve block; pelvic-floor exam

- **Ketamine cystitis** causes HL-like inflammation, fibrosis and **hydronephrosis (~30%)**. Suspect it in the **young (<30) with visible inflammation**, and treat by **cessation**.
- **HTPFD** gives pelvic pressure, obstructive voiding and dyspareunia, with tender **myofascial trigger points**. Missed HTPFD causes poor response, so refer to a pelvic-floor physiotherapist.
- **Urodynamics** are not standard. Compliance is **normal**, and detrusor overactivity **does not exclude IC/BPS**.

Cystoscopy and bladder tests

- **AUA 2022** advises office cystoscopy when **HL is suspected, especially over 50**. Flexible cystoscopy shows most HLs **without hydrodistension**; biopsy **any trigonal lesion**.
- Relief after an **anaesthetic challenge** shows bladder-centred pain and selects intravesical therapy.
- The **potassium sensitivity test** (**0.4 M KCl** vs water) is **not recommended**, because it is painful and non-specific.
- **Hydrodistension** under anaesthesia fills at **80 cm H_2O for 1–3 min** to show **glomerulations**, which are **non-specific**. Biopsy only **after** distension and only suspicious lesions.

8 · PHENOTYPE-DIRECTED MANAGEMENT

P4 *Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management — four-phenotype table: HL, anaesthetic-responsive, pelvic floor, widespread; matched treatment*

> **KEY POINT**
>
> MAPP reanalysis (2023) of two negative NIH RCTs: hydroxyzine/PPS helped **localised pelvic pain**; amitriptyline helped **widespread pain**. A trial negative in all-comers can hide a positive result in the right phenotype.

9 · TREATMENT — AUA 2022 PRINCIPLES

The aim is coping and quality of life, not cure.

- There are **no tiers**, only grouped options: patients may move to any category **except major surgery** without failing "lower" ones. Stop ineffective treatment and seek other pain generators.
- The **Standard** is manual (myofascial) physiotherapy for pelvic-floor tenderness. Fulguration and/or triamcinolone for HLs is a **Recommendation**. Most drugs are **Options**. Chronic opioids are not routinely recommended.

ALGORITHM

1. If symptoms last ≥6 weeks, take a history, examine (pelvic floor, Q-tip, DRE), and check PVR, urinalysis and culture, with cytology if indicated.
2. Exclude confusable disease.
3. If HL is suspected (especially >50), then do cystoscopy. If HL is found, then fulgurate or inject triamcinolone; if refractory, give CyA.
4. If there is no HL, then match the phenotype: pelvic floor, physiotherapy; lidocaine responder, intravesical therapy; widespread pain, systemic therapy and CBT.
5. Combine behavioural, oral and intravesical treatment as needed, then BTX-A or sacral neuromodulation (SNM).
6. Only if bladder-centred and all else fails (rare), consider diversion with or without cystectomy.

10 · TREATMENT OPTIONS (AUA 2022)

P4 *Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management — options table: behavioural, physical (avoid Kegel), oral, intravesical, procedures, refractory HL, last resort*

Not in the paper answer:

- SNM is favoured over BTX-A for HTPFD or obstructive voiding.

10.1 Behavioural and pelvic-floor therapy

- Common **diet triggers** are caffeine, alcohol, fizzy drinks, spicy food and citrus; an elimination diet identifies the patient's own triggers.
- Education and behavioural modification alone improved 45% in the NIH amitriptyline RCT.
- In the myofascial physiotherapy RCT, 10 sessions beat global massage, **59% vs 26%**.

10.2 Oral therapy

- **Amitriptyline**: start **10 mg at night, adding 10 mg weekly to ~50 mg**. In the NIH RCT it gave 55% vs 45% (not significant), but **77% vs 53%** if ≥50 mg was tolerated. **Contraindications** are long QT, conduction block, recent myocardial infarction and heart failure.
- **Hydroxyzine** (H1 blocker) 25–50 mg at night had a negative RCT; **cimetidine** 400 mg twice daily has one positive RCT.
- **PPS 100 mg three times daily** is the only FDA-approved oral IC drug; only **3–6%** reaches urine and it acts in **3–6 months**. Its phase IV RCT was negative.
- **PPS maculopathy** is cumulative-dose related. **FDA (2020)**: retinal examination for all **within 6 months**, then periodically.
- **Ciclosporin A** is for HL refractory to fulguration or triamcinolone, at **1–3 mg/kg/day in two doses**. Its main toxicities are **nephrotoxicity and hypertension**, so monitor blood pressure, renal function and levels.
- **Antibiotics**: the AUA advises **none long-term**, and only one empirical course if never treated.

10.3 Intravesical therapy

Intravesical therapy is for **bladder-centred** pain; a poor lidocaine response predicts failure of every instillation. It is an AUA Option.

Agent	Regimen	Key points
DMSO 50%	50 mL, **15–20 min dwell**, every 1–2 weeks × 4–6	FDA-approved. Better in HL. Causes garlic breath and an initial flare; increases absorption of co-instilled lidocaine.
Heparin	10,000–40,000 U, 30–60 min, self-instilled	GAG mimic; benefit takes months.
Alkalinised lidocaine	200 mg with 8.4% bicarbonate daily × 5	Immediate effect, so useful for flares; tachyphylaxis.

Agent	Regimen	Key points
Hyaluronic acid, chondroitin, PPS	Weekly then monthly	Weak/negative trials

- Cocktails are no better than DMSO alone.

10.4 Hunner lesion treatment

- **AUA Recommendation**: fulguration and/or submucosal triamcinolone, with biopsy if never biopsied. **75–90%** improve, but repeat treatment is usually needed within 1–2 years.

ALGORITHM

1. Under general anaesthesia, map the HLs before distension causes bleeding.
2. Distend mildly and biopsy suspicious or trigonal lesions.
3. Fulgurate at **low volume** (Bugbee electrode), working from the **outer margin inwards**.
4. Inject triamcinolone 40 mg/mL (or 1:1 with bupivacaine/lidocaine) in 0.5-mL aliquots, 2–3 mm submucosally.
5. Fulgurate bleeders, empty the bladder and instil lidocaine–bupivacaine.

KEY POINT

Fulgurate from the margin inward: cautery erythema and oedema spread peripherally, so the true edge is lost if you start centrally.

10.5 Hydrodistension

- AUA Option: **short (1–3 min), low-pressure (60–80 cm H_2O)** distension, with benefit fading by 6 months.
- An anaesthetic capacity **<400 mL** marks a bladder-centred phenotype, often HL.

10.6 OnabotulinumtoxinA and neuromodulation

- **BTX-A** is an Option after failed conservative, oral and intravesical therapy: **100 U** into the bladder including the **trigone**, lasting about 5–9 months. The patient **must accept clean intermittent catheterisation (CIC)**.
- **SNM (S3)** is an Option and works best with **HTPFD or levator tenderness**, but IC/BPS has high explantation rates.

	BTX-A	SNM
Favoured when	Bladder-centred pain; can do CIC	Obstructive voiding, incomplete emptying, HTPFD, CIC impossible
Main risk	Retention needing CIC	Adverse events, explantation

10.7 Major surgery — last resort

- Surgery is only for **rare (<10%), selected, bladder-centred** patients failing all else, and pain may persist after cystectomy. If pain persists after a **differential spinal block** to T10, surgery will fail.

Predicts success	Predicts failure
HLs (refractory)	No HLs
Small fibrotic bladder; anaesthetic capacity <200–400 mL	High anaesthetic capacity
Profound relief with intravesical anaesthetic	Equivocal/no response
Painful filling, pain on scope touch	Urethral/neuropathic pain; HTPFD; widespread pain

- Cystectomy with a conduit leaves about three-quarters pain-free. Prefer the **simplest diversion**, as a retained bladder risks pain and pyocystis.

11 · WHAT NOT TO DO (AUA)

Do not offer	Why
Long-term oral antibiotics	RCT negative; adverse events 80% vs 40%
Intravesical bacillus Calmette–Guérin (BCG)	Ineffective; DMSO superior; † BCG sepsis
High-pressure, long hydrodistension†	No benefit; rupture, sepsis †
Long-term oral glucocorticoids	No benefit; toxicity
† Intravesical resiniferatoxin	Ineffective in RCT †
Kegel strengthening	Worsens HTPFD pain
Major surgery first / non-bladder-centred pain	Irreversible; pain persists

KEY POINT

Nothing is worse than irreversible surgery followed by continued pain — every predictor of surgical success marks bladder-generated pain.

12 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Myofascial PT RCT (FitzGerald 2012)	10 sessions myofascial vs global massage	59% vs 26%	Guideline Standard
NIH amitriptyline RCT (Foster 2010)	EBMP with or without amitriptyline	55% vs 45% (not significant); 77% vs 53% if ≥50 mg	Titrate to ≥50 mg; behavioural therapy works
PPS phase IV RCT	PPS vs placebo	No difference (≈40% each)	Efficacy doubtful; remains Option
MAPP phenotype reanalysis (2023)	Two negative NIH RCTs by pain distribution	Hydroxyzine/PPS localised; amitriptyline widespread	Treat by phenotype

29 Sexually Transmitted Diseases

1 · SCREENING, PREVENTION AND PARTNER NOTIFICATION

One sexually transmitted disease (STD) often hides another.

- The nucleic acid amplification test (**NAAT**) is the standard screen; men who have sex with men (MSM) need urethral, rectal and pharyngeal tests. MCQ Test the pharynx for gonorrhoea, not chlamydia.
- **Partner notification** covers the previous **60 days** for urethritis, chlamydia and Mycoplasma genitalium; all partners for trichomoniasis; and the previous month for scabies or lice. Abstain for 7 days and until partners are treated. Test everyone at risk for HIV and syphilis.

2 · URETHRITIS

- **Gonococcal** urethritis makes up 5–20%. In non-gonococcal urethritis (**NGU**), C. trachomatis is commoner than M. genitalium, which is the leading cause of **persistent or recurrent** urethritis. Up to half have no pathogen.
- **Diagnosis** needs any one: discharge; Gram stain with **≥2 white blood cells (WBC) per oil-immersion field**; positive leucocyte esterase on first-void urine; or ≥10 WBC/hpf in first-void urine.
- Intracellular gram-negative diplococci in a symptomatic man confirm gonorrhoea; a negative Gram stain does not exclude it.

- **First-void urine NAAT** is preferred; **culture** is for treatment failure and children.

Pathogen	First line	Alternatives / notes	Follow-up
Gonorrhoea (CDC 2021)	**Ceftriaxone 500 mg IM once** (1 g if ≥150 kg)	Gentamicin 240 mg IM and azithromycin 2 g (β-lactam allergy); cefixime 800 mg. Europe: ceftriaxone 1 g and azithromycin 2 g. Add chlamydia cover unless excluded	Test of cure only for **pharyngeal** infection; retest at 3 months
Chlamydia / NGU	**Doxycycline 100 mg bd 7 days**	Azithromycin 1 g once	No test of cure; retest at 3 months
M. genitalium	Doxycycline 7 days, then **azithromycin** 1 g, then 500 mg daily × 3, if macrolide-sensitive; or **moxifloxacin** 400 mg × 7 days if resistant or untested	Single-dose azithromycin is inadequate	Abstain until cured
T. vaginalis (male)	Metronidazole 2 g once	Tinidazole 2 g once	Treat all partners

- Higher-dose **ceftriaxone monotherapy** reflects resistance; cephalosporins are safe unless a recent IgE-mediated penicillin reaction occurred. Treatment failure needs culture and susceptibility.
- **M. genitalium** has no cell wall, so Gram stain misses it; NAAT with a macrolide-resistance assay picks the second drug.
- There is no chlamydia test of cure, because nucleic acid persists ~3 weeks.

ALGORITHM

1. Persistent NGU: check adherence and look for an untreated partner.
2. Send NAAT for M. genitalium, with or without T. vaginalis.
3. If negative, then think of chronic pelvic pain syndrome.

KEY POINT

Men with gonorrhoea usually have symptoms; women are usually silent and present with pelvic inflammatory disease or infertility. Treating the man protects his partner.

3 · EPIDIDYMITIS BY AGE

Acute means <6 weeks; chronic (≥6 weeks) is mostly tuberculosis (see Ch 27). Always exclude **torsion**. Apply urethritis criteria and NAAT; add urine culture if enterics are likely.

Group	Organisms	Empiric regimen (10 days)
Young, sexually active	C. trachomatis, N. gonorrhoeae	Ceftriaxone 500 mg IM and doxycycline 100 mg bd
Insertive anal sex	Above, plus enterics	Ceftriaxone and levofloxacin 500 mg daily
>35 years, obstruction or instrumentation	E. coli	Levofloxacin alone

4 · GENITAL ULCER DISEASE

Herpes is commonest, then syphilis. Appearance misleads, so test **every ulcer** with syphilis serology, HSV NAAT, an H. ducreyi test if endemic, and HIV. Biopsy non-healing ulcers, because carcinoma mimics them.

Disease (organism)	Lesion	Nodes	Test	Treatment
Primary syphilis (T. pallidum)	**Painless**, indurated, clean, single	Painless, rubbery, bilateral	Serology; darkfield	Benzathine penicillin G 2.4 MU IM once
Herpes (HSV-2 more than HSV-1)	**Painful** multiple vesicles that become shallow ulcers	Tender, bilateral	NAAT of unroofed base	Aciclovir and related drugs
Chancroid (H. ducreyi)	**Painful, ragged, undermined, purulent**	Tender **suppurative buboes**	Special culture; exclusion	Azithromycin 1 g once, or ceftriaxone 250 mg IM once, or ciprofloxacin 500 mg bd 3 days, or erythromycin 7 days
Lymphogranuloma venereum, LGV (C. trachomatis L1–L3)	Small, painless, transient	Painful, matted, fistulae; groove sign	Chlamydia NAAT	Doxycycline bd **21 days**
Donovanosis (Klebsiella granulomatis)	**Painless, beefy red**, bleeds easily	**Absent** (pseudobuboes)	**Donovan bodies** on crush preparation	Azithromycin 1 g weekly or 500 mg daily, ≥3 weeks **and until healed**

Genital herpes

- HSV lies latent in the **sacral ganglia** and reactivates with trauma, surgery or immunosuppression. Most transmission is from asymptomatic shedding. HSV-2 recurs more, so type it.
- Complications are aseptic meningitis and **urinary retention**; HSV-2 raises HIV risk 2–3-fold.

Drug	First episode (7–10 days)	Suppressive	Episodic
Aciclovir	400 mg tds	400 mg bd	800 mg bd 5 days
Valaciclovir	1 g bd	500 mg–1 g daily	500 mg bd 3 days
Famciclovir	250 mg tds	250 mg bd	1 g bd 1 day

- Treat every first episode. **Episodic** therapy must start within 1 day of lesions or in the prodrome.

Chancroid, donovanosis, LGV

- **Chancroid** needs a painful ulcer, a typical picture, and syphilis and HSV excluded. Aspirate or drain fluctuant buboes.
- **Donovanosis**: if antibiotics fail, biopsy, because **penile squamous cell carcinoma (SCC) mimics it**.

P1 *Lymphogranuloma venereum — L1–L3; proctocolitis in MSM now commonest; groove sign; elephantiasis, rectal stricture; inguinal disease commoner in men; doxycycline 21 days*

5 · SYPHILIS STAGING AND SEROLOGY

Stage	Features	Treatment (penicillin-allergic)
Primary (10–90 days)	Chancre and nodes; heals in 3–8 weeks	Benzathine penicillin **2.4 MU IM once** (doxycycline 14 days)
Secondary (3–5 months)	Rash on **palms and soles**, **condylomata lata**, alopecia, hepatitis, nephritis	As primary
Early latent (<1 year)	Positive serology only	As primary
Late latent / unknown	Positive serology only	Benzathine **weekly × 3** (doxycycline 28 days)

Stage	Features	Treatment (penicillin-allergic)
Tertiary	Gumma, aortitis, tabes dorsalis	Weekly × 3 if cerebrospinal fluid (CSF) normal
Neurosyphilis (any stage)	Central nervous system, eye, ear	**Aqueous penicillin G 18–24 MU/day IV 10–14 days**

- T. pallidum cannot be cultured, so **two serological tests** are used. The **nontreponemal test (NTT)** (VDRL, RPR) can be falsely positive; its titre falls with treatment, so it **monitors treatment**. The **treponemal test (TT)** (FTA-ABS, TP-PA, EIA) usually stays positive for life.
- A significant change is **fourfold (two dilutions)**. Use the same NTT each time.

ALGORITHM

1. **Reverse algorithm**
2. Start with a TT. If positive, then do an NTT.
3. If the NTT is negative, then do a second, different TT.
4. If that is positive and there is no past treatment, then treat as late latent syphilis.

- The **Jarisch–Herxheimer** reaction (fever and myalgia within 24 h) is treponemal lysis, not allergy; give NSAIDs.
- In primary or secondary syphilis, test for HIV and offer pre-exposure prophylaxis (PrEP) if negative.
- **Follow-up** serology is at 6 and 12 months (and 24 for latent disease). A fourfold rise means failure or reinfection, so retreat weekly × 3 and consider CSF.

6 · HPV AND ANOGENITAL WARTS

- Human papillomavirus (HPV) **6 and 11** cause ~90% of warts. HPV **16 and 18** cause >70% of cervical cancer, and also penile and anal cancer.
- **Bowenoid papulosis** (HPV-16) is a high-grade squamous intraepithelial lesion. **Giant condyloma** (Buschke–Löwenstein) is a locally invasive verrucous carcinoma.
- Diagnosis is clinical. **Biopsy if** the diagnosis is uncertain, there is no response or worsening, the patient is immunocompromised, or the lesion is pigmented, indurated, fixed or ulcerated.

Treatment	Key points
Imiquimod 5% (patient)	3 nights/week for up to 16 weeks; wash off after 6–10 h. Weakens condoms
Podofilox 0.5% (patient)	bd for 3 days, then 4 days off. **Contraindicated in pregnancy**
Sinecatechins 15% (patient)	tds, not washed off. Weakens condoms. Not in HIV, herpes or pregnancy
Cryotherapy, trichloroacetic acid (TCA) 80–90% (provider)	Cryotherapy every 1–2 weeks; TCA weekly
Excision, curettage, CO_2 laser	For large warts; use smoke extraction

Change treatment if there is no response by 3 months; observation is acceptable.

- The **vaccine** is **nonavalent** (6, 11, 16, 18, 31, 33, 45, 52, 58). Give it at **11–12 years** (from 9), with catch-up to **26**, in both sexes. Starting <15 needs **2 doses**; ≥15 or immunocompromised needs **3 doses**.

7 · MOLLUSCUM AND ECTOPARASITES

- **Molluscum** contagiosum forms **umbilicated** papules that regress in 6–12 months; use cryotherapy, curettage or imiquimod. Extensive lesions suggest HIV.
- **Pubic lice**: use a **permethrin 1%** rinse for 10 min, or malathion 0.5% or ivermectin 250 µg/kg if resistant.

- **Scabies** causes nocturnal itch and burrows. Apply **permethrin 5%** to all skin, or give ivermectin 200 µg/kg repeated at 2 weeks. **Crusted** scabies (immunocompromised) needs topical treatment and repeated ivermectin, never lindane.
- Treat contacts from the past month and wash bedding.

8 · HIV FOR UROLOGISTS

P1 *Describe clinical implication of AIDS on the Genito-Urinary Tract? — staging (AIDS = CD4 <200), diagnosis, GU manifestations by site (UTI, prostate, testis, voiding, ED, HIVAN, indinavir stones, cancers),*
PEP 72 h/28 days, needlestick 0.3%, circumcision 50–60%

Not in the paper answer:

- Start post-exposure prophylaxis (PEP) within **72 h**, and only for mucosal, broken-skin or percutaneous contact with blood or genital fluid. Needlestick risk is **0.3%**.
- Circumcision cuts heterosexual acquisition by **50–60%** but does not protect partners. **After sexual assault** (men), give ceftriaxone 500 mg once, doxycycline 7 days, hepatitis B and HPV vaccines, and PEP within 72 h.
- PrEP is daily emtricitabine with tenofovir disoproxil fumarate, or tenofovir alafenamide in men only.

30 Cutaneous Diseases of the External Genitalia

1 · APPROACH TO GENITAL DERMATOSES

Genital rashes look alike, so biopsy early to avoid missing cancer.

- Genital skin has few reaction patterns, so mimics abound and the **biopsy threshold is low**. Ask about over-the-counter creams, and do a **full skin survey**, because extragenital lesions often give the diagnosis.
- Identify the **primary lesion**. A papule is solid and **≤0.5 cm**; larger solid lesions are nodules or plaques, and fluid-filled ones are vesicles (≤0.5 cm) or bullae. An **erosion** loses epidermis only and does not scar; an **ulcer** loses dermis too and scars.
- Bedside tests: KOH prep for hyphae or yeast, mineral oil prep for scabies, Tzanck smear for herpes, and Wood's lamp for erythrasma (coral-red) and vitiligo.

Biopsy: a 2–3 mm punch under local anaesthetic is often diagnostic; large or meatal lesions go to theatre.

2 · PRINCIPLES OF THERAPY

- Use **topical treatment first**; urologists overprescribe systemic antibiotics.
- Give **systemic antifungals** only for extensive dermatophytosis, Majocchi granuloma, severe immunocompromise or topical failure, and avoid systemic **ketoconazole** (hepatotoxicity, adrenal insufficiency).
- Give systemic prednisone for ≤3 weeks as a single morning dose.
- **Topical steroids** cause atrophy in thin genital skin, and the **prepuce acts as an occlusive dressing**. The rule is low potency and a short course; lichen sclerosus (LS) is the exception.
- Narrow-band UVB or the 308-nm excimer treats psoriasis and vitiligo. **PUVA causes genital squamous cell carcinoma (SCC)**, so it is avoided.

3 · ECZEMATOUS DERMATOSES

Exclude Candida, herpes, impetigo and drug reaction first.

Atopic dermatitis (AD)

- AD is relapsing, flexural and intensely pruritic; **90%** present before 5 years. Use emollients, a short topical steroid, calcineurin inhibitors and a sedating antihistamine; severe disease needs **dupilumab**.

Contact dermatitis

Feature	Irritant (ICD)	Allergic (ACD)
Mechanism	Direct cytotoxicity; dose-driven	Type IV hypersensitivity
Share	~**80%**	Minority; genital ACD is rare
Agents	Soaps, solvents, acids/alkalis	**Nickel** (commonest), neomycin, fragrances, latex, topical steroids
Picture	Burning, limited to the exposed zone	Demarcated, pruritic, weeping
Confirm	Exposure history	**Patch testing**
Treat	Avoidance, barrier	Remove allergen. If severe, a **3-week** prednisone taper (a short course rebounds)

Seborrhoeic dermatitis (SD)

- SD gives greasy-scaled plaques in sebaceous areas. It affects up to **83%** of patients with AIDS, so **extensive SD should prompt an HIV test**. A topical **antifungal** (ketoconazole) is the backbone.
- Lichen simplex chronicus (lichenification, in the LS differential) needs the itch–scratch cycle broken with a short potent steroid.

4 · PAPULOSQUAMOUS DISORDERS

Tinea, erythrasma, secondary syphilis and neoplasia (Bowen disease, extramammary Paget disease (EPD), mycosis fungoides) mimic psoriasis, so "psoriasis" failing treatment needs KOH, syphilis serology and biopsy.

Psoriasis

- Psoriasis affects ~2% and forms sharp plaques with silvery scale; check nails, scalp, **umbilicus** and natal cleft. **Inverse psoriasis** loses its scale in folds, so use KOH to exclude tinea or Candida.

Step	Genital psoriasis treatment
First-line	Low-potency steroid (hydrocortisone butyrate 0.1%) continuously for **≤2 weeks**; triamcinolone 0.1% for thick plaque
Steroid-sparing	Calcipotriene, calcineurin inhibitors
Localised	308-nm excimer laser
Contraindicated	**PUVA** (dose-dependent genital SCC)
Extensive	Methotrexate, cyclosporine, apremilast, anti-TNF, ustekinumab, **ixekizumab (anti-IL-17), which has the best genital data**

Reactive arthritis (formerly Reiter)

- Reactive arthritis follows urethritis or enteric infection in **HLA-B27**-positive people. **Balanitis circinata** looks like psoriasis, so diagnose from **context**; it is self-limiting and needs only a low-potency steroid.

Lichen planus (LP)

- LP forms **polygonal, violaceous, flat-topped papules**; glans papules are white and reticular. Papular LP usually resolves within a year.
- **Erosive mucosal LP persists, scars and carries an SCC risk**, so review regularly and biopsy ulcers. Use an **ultrapotent** steroid (clobetasol 0.05%).

5 · LICHEN SCLEROSUS (LS) / BXO

P1 *Lichen sclerosus (BXO, balanitis xerotica obliterans) — scarring disease; phimosis, then meatus, then anterior stricture; histology; HPV-independent SCC 2.3–9%; clobetasol 3 months, then maintenance; circumcision; BMG not genital skin*

Not in the paper answer:

- LS is 6–10× commoner in females, and affected men are usually >60; **15–20%** have extragenital disease, and in men the **perianal skin is spared**.
- Circumcision is usually curative, but LS may recur in the scar.

ALGORITHM

1. Biopsy.
2. Give a **potent steroid (clobetasol 0.05%) for 3 months**, then taper.
3. Continue twice-weekly maintenance (or mometasone 0.1% or a calcineurin inhibitor).
4. Lifelong surveillance of non-healing fissures, ulcers, nodules.

- Recurrence is **40–50%**, so maintain proactively. Meatal disease needs meatotomy or meatoplasty.

Disease	Steroid potency	Duration
AD / psoriasis / reactive arthritis	Low	Short (psoriasis ≤2 weeks)
Seborrhoeic dermatitis	Low	Initial severe phase only
Lichen planus	Ultrapotent	Until controlled
Lichen sclerosus	Potent	3 months, then maintenance

KEY POINT

Steroid strength tracks the scarring and malignant stakes of the disease. LS is the single exception to the short-course rule.

6 · VESICULOBULLOUS AND DRUG ERUPTIONS

Feature	Erythema multiforme (EM) minor	Stevens–Johnson syndrome (SJS)	Fixed drug eruption (FDE)
Trigger	**HSV**	**Drugs**: sulfonamides, NSAIDs, β-lactams, anticonvulsants	Sulfonamides, NSAIDs, tetracyclines, carbamazepine
Lesion	Symmetric **targets**; palms, soles, mouth	Blisters and necrosis; genitals involved in ~**60%**	Solitary violaceous plaque, often on the glans
Course	Self-limited, recurs	Mortality up to ~**30%**	**Recurs at the same site within 24 h** of re-exposure
Sequelae	None	Meatal stenosis, strictures	Brown pigmentation
Treatment	**Suppressive aciclovir prevents it**	See algorithm	Stop the drug

ALGORITHM

1. **SJS**
2. Stop the offending drug.
3. Give burn-type supportive care in an intensive care or burn unit (lowers mortality).
4. Limited data support cyclosporine 3–5 mg/kg/day, etanercept, plasmapheresis or IVIG; systemic steroids are controversial.

KEY POINT

One drug class can cause a trivial FDE or a lethal SJS; extent and systemic illness separate them. Genital FDE mimics herpes; "same spot each time" is the clue.

Autoimmune blistering disease needs perilesional biopsy with direct immunofluorescence.

7 · NONINFECTIOUS GENITAL ULCERS

A genital ulcer is infectious until proven otherwise (see Ch 29).

Behçet disease (BD)

- BD is an **HLA-B51**-linked vasculitis, **more severe in males**, with oral and genital ulcers, uveitis, epididymitis and **pulmonary artery aneurysm**. Oral ulcer is most sensitive; **genital ulcer is most specific**. No confirmatory test exists. Treat with colchicine, steroids or **anti-TNF**.

International Criteria for BD (2014) — ≥4 points	Points
Ocular lesions; oral aphthosis; genital aphthosis	2 each
Skin lesions; central nervous system; vascular; positive pathergy (optional)	1 each

Pyoderma gangrenosum (PG)

- PG accompanies inflammatory bowel disease and myeloproliferative disease as a painful ulcer with an **undermined violaceous border**. Biopsy the edge, treat with steroids, and avoid debridement.

8 · INFECTIONS

Balanitis and balanoposthitis

ALGORITHM

1. Remove irritants and improve hygiene.
2. Give a topical antibiotic or antifungal, then a short low-potency steroid.
3. If this fails, then **biopsy** (neoplasia, Zoon, psoriasis, HPV, HSV).
4. If recurrent with phimosis, then **circumcision**.

Type	Clue	Management
Candidal	Satellite lesions; diabetes	Topical imidazole; check glucose
Irritant/allergic	Products, soaps	Avoidance
Circinate	Urethritis, HLA-B27	Low-potency steroid
Zoon (plasma cell)	Shiny red plaque, elderly uncircumcised	Biopsy, then circumcision
LS / BXO	White sclerotic glans, phimosis	Potent steroid; circumcision
Pseudoepitheliomatous keratotic micaceous balanitis (PEKMB)	Mica-like scaly plaque	Biopsy; excision

Soft-tissue infection

- Folliculitis needs hygiene. A furuncle needs incision and drainage with MRSA cover. Erysipelas has a sharp border and cellulitis an indistinct one; give antibiotics and **mark the border**.
- Fournier gangrene is a polymicrobial fascial infection with pain out of proportion, grey skin and crepitus; it needs emergency debridement (see Ch 26).

KEY POINT

Fournier gangrene is fascial, so skin shows a fraction of the necrosis; testes are spared. Imaging must never delay surgery. Debride to bleeding tissue, with a second look at 24–48 h.

- **Ecthyma gangrenosum** (P. aeruginosa bacteraemia in the immunocompromised) gives black eschar and needs immediate IV antipseudomonals.
- A **genital bite** needs amoxicillin–clavulanate 875/125 mg bd, irrigation and debridement.

Hidradenitis suppurativa (HS)

- HS is a **follicle-centred** disease of apocrine-bearing skin that causes abscesses, **sinus tracts** and scars, with fistulae and **SCC in chronic scars** as complications.

Hurley stage	Features	Mainstay
I	Abscess(es), no tracts or scarring	Topical clindamycin, deroofing
II	Recurrent abscesses, tracts, scarring; separated areas	Oral tetracycline, or clindamycin and rifampicin; deroofing; adalimumab if refractory
III	Diffuse interconnected tracts	Adalimumab and **wide excision**, graft or flap

- Advise stopping smoking and weight loss; **adalimumab** is given 40 mg weekly. **Repeated incision and drainage is discouraged** because the tract recurs.

Red groin rash

Feature	Candidal intertrigo	Tinea cruris	Erythrasma	Inverse psoriasis
Scrotum	**Involved**	**Spared**	—	May be involved
Look	**Satellite lesions**	Scaly advancing border, central clearing	Red-brown patch	Non-scaling plaque
Test	KOH: yeast	KOH: hyphae	Wood's lamp: **coral-red**	KOH negative
Treat	Topical imidazole **≥2 weeks**; test for diabetes	Topical azole or allylamine; terbinafine 250 mg/day if deep; no steroids	Clindamycin 1% or miconazole; oral erythromycin if severe	Low-potency steroid

> **KEY POINT**
> Azoles cover yeast and dermatophyte, so use them when unsure. Nystatin misses tinea; allylamines are weaker on Candida. A negative KOH from an untreated lesion means rethink the diagnosis.

Infestations

- Pubic lice and scabies both get permethrin 5%, **repeated at 1 week**; for scabies treat the whole body and the household. Ivermectin 200 µg/kg ×2 is the alternative, and lindane is disfavoured. Screen lice patients for other STDs.

9 · PREMALIGNANT AND MALIGNANT LESIONS

P1 *Premalignant conditions in urology (including premalignant lesions of penis) — PeIN HPV vs non-HPV; Queyrat, Bowen, bowenoid papulosis; Buschke–Löwenstein (no RT); glans therapy (5-FU, imiquimod,*
laser, resurfacing)

Not in the paper answer:

- Penile intraepithelial neoplasia is HPV-related (type 16) or non-HPV (LS, chronic LP); biopsy **several areas**.
- Verrucous carcinoma needs local excision or Mohs surgery, not primary radiotherapy (anaplastic transformation).

Lesion	Treatment
SCC in situ (SCCis) — accessible (scrotum)	Excision with a **5-mm margin** (favoured)
SCCis — glans	Mohs surgery, laser, cryotherapy; topical 5-fluorouracil (5-FU) or imiquimod (higher recurrence)

Lesion	Treatment
Bowenoid papulosis	Observe in a reliable young man; or 5-FU, imiquimod, cryotherapy, laser
PEKMB	Biopsy (exclude verrucous carcinoma); excision or ablation

Tumour	Key features	Treatment
Basal cell carcinoma	Rare; pearly papule	Excision or Mohs
Kaposi sarcoma	HHV-8; classic, AIDS and transplant types; a glans lesion may block the meatus	Restore immunity; excision, laser or vinblastine if solitary; radiotherapy 15–30 Gy if regional; chemotherapy if disseminated
Melanoma	<1% of penile cancers	Wide excision or partial penectomy; immunotherapy if advanced
Extramammary Paget	Red plaque at apocrine sites; CEA and EMA positive; underlying carcinoma in **10–30%** (urethra, prostate, bladder, rectum)	**Mandatory search for internal malignancy**; excision or Mohs
Mycosis fungoides	Patches progress to plaques, then tumours; Sézary syndrome if in blood	Ultrapotent steroid, phototherapy

10 · BENIGN MALE GENITAL LESIONS

Lesion	Site / look	Management
Pearly penile papules	Coronal rows of angiofibromas in 14–48% of young men, mistaken for condyloma	Reassure
Fordyce spots	Pin-sized shaft papules (ectopic sebaceous glands)	Reassure
Angiokeratoma of Fordyce	Red-purple scrotal papules; may bleed	Observe; laser if bleeding
Zoon balanitis	Shiny red glans plaque; elderly uncircumcised men	**Biopsy**, then **circumcision (curative)**
Median raphe cyst	Ventral, near the glans	Excision

KEY POINT

Zoon balanitis and erythroplasia of Queyrat are both red glans plaques in uncircumcised men: one benign and cured by circumcision, the other SCCis. A persistent red glans plaque always needs biopsy.

11 · PIGMENTARY DISORDERS

- Vitiligo is accentuated by Wood's lamp; unlike LS, it has normal texture without atrophy or scarring.

31 Tuberculosis and Parasitic Infections of the Genitourinary Tract

1 · UROGENITAL TUBERCULOSIS — ORGANISM AND PATHOGENESIS

TB seeds the urinary tract late and heals by scarring.

P3 P2 *Renal and genitourinary TB: etiopathogenesis, presentation, diagnosis and management — MTBC, granuloma/IFN-γ, latency and reactivation, routes, renal sequence to autonephrectomy, site frequencies*

Not in the paper answer:

- UG TB is **15–20%** of pulmonary TB in developing countries. Infection stays latent in 90–95%; lifetime reactivation is **5–15%**, so UG TB often appears without active lung disease.
- **M. bovis (and bacille Calmette–Guérin, BCG) is innately pyrazinamide-resistant**.
- Isolated urogenital (UG) TB is generally **not infectious**, except during aerosol-generating procedures.
- Ascending spread: intravesical BCG causes UG TB in **~1%**.

> **KEY POINT**
> UG TB is blood-borne and lands in the two best-perfused UG organs, kidney and epididymis. Blood-borne seeding gives parenchymal disease; descending urinary spread gives luminal disease (distal ureteric stricture, cystitis). Healing by fibrosis turns an infection into an obstruction.

2 · CLINICAL FEATURES AND PATHOLOGY BY SITE

- It is indolent, and patients are repeatedly treated for urinary tract infection (UTI). Constitutional symptoms occur in **<20%**. **Lower urinary tract symptoms dominate**; haematuria or flank pain occurs in ~⅓. **The hallmark is sterile pyuria** (>90%).

Site	Frequency / route	Signature lesion
Kidney	**80%**, haematogenous	Caseating cavities, then autonephrectomy (up to ⅓); "putty" kidney; end-stage renal failure ~7%
Ureter	Descending	Stricture at the vesicoureteric junction (VUJ); beaded corkscrew ureter. **Stricture obstruction is an important cause of renal failure**
Bladder	Descending	Starts at orifices; trigone spared. After ~1 year <100 mL; "thimble" <20 mL
Epididymis	10–55%; second commonest; bilateral in ~⅓	Starts in the **globus minor (tail)**; posterior scrotal sinus in up to half
Vas	Contiguous	Thickened, beaded
Testis	From epididymis	Hard tumour-like mass
Prostate	22–49%	Calcified hard gland. "Prostatitis" responds partly to quinolones, then relapses
Seminal vesicles	Via vas or ejaculatory ducts	Low-volume ejaculate, azoospermia, haemospermia; infertility may be the first symptom
Urethra	1.9–4.5%	With prostatic disease; urethroscrotal fistula
Penis	Exceedingly rare	Ulcer; papulonecrotic tuberculid

3 · DIAGNOSIS

P3 *Renal and genitourinary TB: etiopathogenesis, presentation, (laboratory and newer) diagnosis and management — 3–5 early-morning urines, ZN, LJ/MGIT culture (gold standard, monitors treatment),*
GeneXpert, LF-LAM, TST/IGRA, imaging signs, cystoscopy, algorithm

Not in the paper answer:

- KUB calcification progresses from punctate to globular to ring, and IVU may show cloverleaf calyces.

ALGORITHM

1. **Hierarchy of proof**
2. If culture or MTBC DNA is positive, then the diagnosis is made.
3. If negative, then caseating granulomas on tissue in the right context confirm it.
4. If still negative, then a compatible picture, exposure and response to empirical treatment suffice.
5. Always look for coexisting pulmonary TB (up to one-fifth): chest radiograph in every patient.

- Culture sensitivity is 10–90% because shedding is intermittent. The Ziehl–Neelsen (ZN) smear detects **<50%**. Acidic urine inhibits mycobacteria, so culture promptly.
- **GeneXpert** detects rifampicin resistance, a surrogate for multidrug-resistant (MDR) TB. The WHO uses it in the initial work-up of all TB, but a negative result does not exclude TB, and it **cannot monitor treatment** because dead bacilli shed DNA.
- Fine-needle aspiration cytology of an epididymal nodule is diagnostic in ~⅔. At cystoscopy ulcers mimic cancer, and a **"golf-hole" orifice** means imaging the upper tract.
- Tuberculin skin test (**TST**) cut-offs are **≥5 mm** for contacts and the immunosuppressed, **≥10 mm** for immigrants, high-risk workers and drug injectors, and **≥15 mm** with no risk factors. The interferon-γ release assay (**IGRA**) has no BCG false positives and is preferred for people from endemic countries. Both show exposure only.

Modality	Signs / role
Plain film (KUB)	Calcification in 24–44%; **lobar "putty" kidney** is pathognomonic of end-stage disease; ureteric calcification is **intraluminal** (schistosomiasis: mural)
Intravenous urography (IVU)	**Moth-eaten calyx**; **phantom calyx**; **Kerr's kink**; **pipe-stem ureter**; contracted bladder. Findings are mostly **obstructive**
CT urography	Standard; granuloma may mimic renal cell carcinoma
Ultrasound (US)	**Serial monitoring of hydronephrosis during treatment**
MRI	Diffusion-weighted imaging separates pyonephrosis from hydronephrosis

KEY POINT

Imaging signs are fibrosis made visible: destruction (moth-eaten calyx, cavities) heals by scarring (phantom calyx, Kerr's kink, pipe-stem ureter, golf-hole orifice, thimble bladder). So obstruction is the commonest finding and must be watched during treatment.

4 · MEDICAL TREATMENT

- Combine drugs to prevent resistance. Check baseline liver function, renal function and HIV status, and use directly observed therapy (DOT).

ALGORITHM

1. **Standard 6-month short course (2HRZE / 4HR)**
2. Intensive, 2 months: daily INH (H), rifampicin (R), pyrazinamide (Z) and ethambutol (E); stop E once fully susceptible.
3. Continuation, 4 months: HR daily (or thrice weekly by DOT).
4. Give pyridoxine throughout.

Drug	Daily adult dose	Signature toxicity	Monitor / note
INH	5 mg/kg (300 mg)	Neuropathy, hepatitis	Pyridoxine 25–50 mg
Rifampicin	10 mg/kg (600 mg)	Hepatitis, **drug interactions**	Antiretroviral and immunosuppressant levels; rifabutin with antiretrovirals
Pyrazinamide	Weight-banded 1000–2000 mg	Hyperuricaemia, hepatitis	M. bovis is resistant

Drug	Daily adult dose	Signature toxicity	Monitor / note
Ethambutol	Weight-banded 800–1600 mg	Optic neuritis (red–green vision)	Check vision; the **only non-hepatotoxic** first-line drug

- Some give 12–24 months for complicated disease, as UG TB relapses more.
- **Monitor** liver function, vision, urine culture and **serial US**, because healing fibrosis can worsen strictures and bladder contraction.
- **Steroids** are anecdotal in UG TB; consider them if a stricture worsens on treatment.

5 · SURGICAL TREATMENT

*About half need surgery, now mostly **reconstructive**.*

- The goal is organ preservation. Indications are obstruction or abscess, a non-functioning infected kidney, resistant hypertension and reconstruction.
- **Timing: 4–6 weeks after starting chemotherapy**, except in emergencies.

ALGORITHM

1. If uraemia or sepsis, then decompress urgently.
2. Try a retrograde double-J stent first (preferred; succeeds in < half, 41%).
3. If that fails, then try an antegrade stent.
4. If that fails, then percutaneous nephrostomy (PCN); loculated systems may need several.
5. Then repair definitively; if the kidney is unsalvageable, then nephrectomy.

- Removing a PCN without fixing the cause leaves a tuberculous cutaneous fistula.
- **Nephrectomy is established** for a non-functioning kidney with recalcitrant TB or resistant hypertension; **routine removal** of an asymptomatic non-functioning kidney is **contested**. Partial nephrectomy suits a polar lesion unresponsive to drugs.
- At nephrectomy, expect dense fibrosis and **ligate artery and vein separately** (late arteriovenous fistula).
- **Strictures** are commonest in the lower ureter, and endoscopic results are poor. Stent early for strictures arising on treatment and **image every stricture**; if worsening, give steroids, and **if there is no improvement after 6 weeks, operate**.

Site / gap	Procedure	Key point
Pelviureteric junction (PUJ), short	Dismembered pyeloplasty; flap if longer	Harder than congenital PUJ obstruction
PUJ unreconstructable	Ureterocalicostomy (lower pole)	Cover with capsule or omentum
Upper/mid ureter	Ureteroureterostomy; Davis intubated ureterotomy	—
Lower ureter	Excise, then ureteroneocystostomy	Bladder mobilisation gains 2–3 cm
Moderate gap	Psoas hitch	Up to **5 cm**; needs good capacity
Long gap	Boari flap, with or without hitch	**10–15 cm**; poor in a contracted bladder
Multiple / recurrent	**Ileal ureter**	Native ureter unusable

KEY POINT

Trap: in TB the bladder may itself be contracted, ruling out psoas hitch and Boari flap. Assess capacity first; ileal ureter, with or without augmentation, is then the option.

P2 *Describe pathophysiology of genitourinary tuberculosis and management of thimble bladder in 30- year male — assessment, augmentation vs substitution, ureteric reimplantation, bowel complications, follow-up*

Not in the paper answer:

- Contracted bladder: operate for intolerable symptoms or capacity **<100 mL**, once chemotherapy controls disease; a thimble bladder needs orthotopic substitution, and the spared trigone allows supratrigonal substitution.
- **Genital** surgery is only for failed medical therapy: **epididymectomy** if the testis is spared, otherwise orchidectomy. Obstructive azoospermia is often irreversible, so offer sperm retrieval.

6 · BCG-RELATED INFECTION, RELAPSE, SPECIAL SITUATIONS

- **BCG UG TB** (~1%) causes granulomatous prostatitis, epididymo-orchitis, renal abscess or sepsis. For organ infection give INH and rifampicin for 3–6 months, since **pyrazinamide is useless**; for sepsis give triple therapy and steroids, and stop BCG permanently.
- **Relapse** occurs in **6.3–22%**, at a mean of ~5 years, so follow up for **10 years**.

Situation	Change	Duration
MDR-TB (INH and R resistant)	Second-line; BPaL (bedaquiline, pretomanid, linezolid), not yet approved for extrapulmonary TB	> 18 months; 6–9 months with BPaL
Extensively drug-resistant (XDR) TB	MDR plus fluoroquinolone and bedaquiline/linezolid resistance; 5–6 drugs	≥ 2 years
Pregnancy	HRE and pyridoxine; Z omitted in the USA (WHO includes it); breastfeeding allowed	9 months (without Z)
HIV on effective antiretroviral therapy (ART)	Standard daily regimen; rifabutin with protease inhibitors	6 months (9 if not on ART)
Renal transplant	Rifamycin with drug-level monitoring, or a rifamycin-free regimen	≥ 18 months if rifamycin-free

- **HIV** raises TB risk 15–20-fold, with more dissemination but **less fibrosis and fewer strictures**.

7 · SCHISTOSOMIASIS (BILHARZIA)

Trapped eggs cause obstruction and squamous bladder cancer.

Species	Distribution	Target / venous home	Snail
S. haematobium	Africa, Arabian Peninsula	Vesical/pelvic plexus — UG tract	**Bulinus**
S. mansoni	Africa, Middle East, S America	Mesenteric — liver, gut	Biomphalaria
S. japonicum	China, SE Asia	Mesenteric — liver, gut	Oncomelania

ALGORITHM

1. **Life cycle**
2. Cercariae penetrate intact skin in fresh water and pass via the lungs to the liver, where they pair.
3. Paired adults move to the vesical plexus and live 3–5 years.
4. Eggs (**terminal spine**) cross the bladder wall into urine and hatch as miracidia.
5. Miracidia infect Bulinus, which releases cercariae.

- **Retained eggs cause the disease**; adults do not multiply in humans. Children aged 5–15 carry the heaviest burden.
- The **egg granuloma** forms polypoid masses (active), then fibroses and calcifies into **"sandy patches"**, leading to obstruction, metaplasia and cancer. Obstruction is **distal**; hydroureter may be **tonic** (reversible once relieved) or **atonic** (fibrotic).
- **Salmonella** shelters in the worm, causing chronic bacteriuria; treat both.
- **Bladder cancer** is rated **Group 1** by IARC. Patients are younger (40–50), and **SCC is 60–90%**; mass drug treatment in Egypt shifted SCC to urothelial cancer.

- Acute infection gives swimmer's itch and **Katayama fever**; chronic infection (from 8–12 weeks) gives **painless terminal haematuria**. Men may get haemospermia and infertility.

> **KEY POINT**
> Trap: a negative urine egg count does not exclude severe disease; the most damaged kidneys belong to egg-negative, often asymptomatic, inactive-stage patients.

Test	Meaning	Limitation
Urine egg microscopy — **gold standard**	Active infection	Collect **09:00–15:00** (noon peak); negative early and in light or inactive disease
Rectal snip (squash prep)	Active or past infection; egg viability	Avoids bladder biopsy risks
Serology	Exposure	Stays positive, so it **never judges cure**
Circulating antigen	Live worms	Urine CCA is poor for S. haematobium
PCR	Most sensitive	Impractical in the field

- Imaging shows wall thickening, polyps and hydronephrosis on US, and a **calcified bladder ("fetal head in the pelvis")** with mural ureteric calcification on plain film.
- **Praziquantel (PZQ)** is the only WHO-listed drug: **40 mg/kg once**, or 20 mg/kg twice 6–8 h apart, curing 60–90%. PZQ is weak on immature worms, so repeat after several weeks.
- Active polypoid obstruction resolves with PZQ alone; late fibrotic stenosis needs surgery.

Complication	Surgery
Distal / intramural stenosis	Antireflux reimplantation; balloon dilatation (restenosis common); meatoplasty for meatal stenosis
Long / multifocal stenosis	Boari flap; ileal ureter; ileal conduit; long-term nephrostomy
Contracted bladder	Ileocystoplasty or diversion, always with PZQ
Chronic deep ulcer	Partial cystectomy (fulguration fails)

8 · FILARIASIS

In India, filariasis commonly causes hydrocele and chyluria.

- W. bancrofti causes >90% and is the **only cause of genital filariasis**; Brugia affects the leg. Microfilariae are **nocturnally periodic**, and doxycycline kills the Wolbachia endosymbiont.
- **Acute adenolymphangitis** is a **retrograde** lymphangitis. **Funiculoepididymitis** is a recurrent painful cord swelling that spares the testis and is often mistaken for tumour.
- **Hydrocele** is the commonest chronic lesion; elephantiasis comes late.
- Diagnosis uses a midnight smear, the **circulating antigen** card, and the US "**filarial dance sign**".

P3 *Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria — pyelolymphatic fistula; ether test, triglycerides, cystoscopic lateralisation, lymphoscintigraphy; MCT diet + DEC, then pelvic instillation sclerotherapy, then hilar lymphatic disconnection*

Not in the paper answer:

- Most chyluria remits on conservative treatment; instil sclerosant on the side identified at cystoscopy.

Drug	Action	Regimen / caution
Diethylcarbamazine (DEC)	Microfilaricidal, modest on adults	Drug of choice: 6 mg/kg/day; a single-day course equals a 12-day course. **Contraindicated if onchocerciasis or Loa loa is possible** (blindness, encephalopathy)
Albendazole	Micro- and macrofilaricidal	400 mg bd for 21 days; single dose in mass treatment

Drug	Action	Regimen / caution
Ivermectin	Microfilaricidal	150–400 µg/kg once; caution with Loa
Doxycycline	Anti-Wolbachia; kills adults	200 mg daily for 4–8 weeks; improves lymphoedema and hydrocele; not in pregnancy or children

- Mass treatment uses albendazole with DEC in India, or with ivermectin where Loa or onchocerciasis is co-endemic.
- **Hydrocele** recurs after aspiration, so **excise the sac**. In funiculoepididymitis excise nodules and preserve the testis. Elephantiasis needs reduction scrotoplasty.
- **Chyluria** occurs when dilated retroperitoneal lymphatics rupture into the collecting system, giving milky urine with fat and protein loss. Exclude TB, tumour and trauma.

9 · HYDATID DISEASE AND OTHER PARASITES

Never puncture a hydatid cyst blindly.

P1 *Hydatid disease of kidney — kidney third site (2–3%); hydatiduria; water-lily sign, WHO stages; albendazole 400 mg bd 1–6 months; no aspiration; scolicide only if non-communicating; nephron-sparing surgery; PAIR not for kidney*

Not in the paper answer:

- Pack the field with hypertonic (20%) saline swabs, and have the anaesthetist ready for anaphylaxis.
- Remove the cyst intact (pericystectomy) or do a partial nephrectomy; nephrectomy is only for a destroyed kidney.
- Enterobiasis, amoebiasis, trichomoniasis (see Ch 29) and fungal UTI (see Ch 26) are rarer genitourinary infections.

★ Putting it together: infection decisions

A · UTI: DRUG MATCHES SITE

Situation	Choice	Principle
Uncomplicated cystitis	Nitrofurantoin 5 days, fosfomycin 3 g once, TMP–SMX 3 days (resistance <20%), pivmecillinam	Narrow, urine-confined
Pyelonephritis, complicated UTI, men	Quinolone or cephalosporin; carbapenem if resistant	Nitrofurantoin and fosfomycin do not reach tissue
ESBL producers	Fosfomycin or nitrofurantoin for cystitis; carbapenem if invasive	Co-resistance to quinolones
Asymptomatic bacteriuria	Treat only in pregnancy and before mucosa-breaching surgery	Otherwise harmful

B · BEYOND ANTIBIOTICS

ALGORITHM

1. **Image**: no response by 72 h, sepsis, obstruction, gas, febrile men.
2. **Drain** any obstructed infected kidney; treat the stone later.
3. Renal abscess: if ≤3 cm, then antibiotics; 3–5 cm, variable; if >5 cm, then percutaneous drainage.
4. **Emphysematous pyelonephritis**: gas within kidney: antibiotics and drainage; if beyond it, then nephrectomy (see Ch 26).
5. **Fournier: urgent repeated radical debridement**.

C · RECURRENT UTI IN WOMEN

ALGORITHM

1. Confirm each episode by culture.
2. Behaviour; vaginal oestrogen if postmenopausal.
3. Methenamine, cranberry, D-mannose.
4. Self-start or post-coital antibiotic; finally continuous prophylaxis.

D · PELVIC PAIN SYNDROMES COMPARED

	CP/CPPS	IC/BPS
Core	Male pelvic pain >3 months, no infection (NIH III)	Filling-related bladder pain, no other cause
Phenotyping	UPOINT; NIH-CPSI	Hunner vs non-Hunner
Treatment	By domain: α-blocker, pelvic-floor physiotherapy, neuromodulators, psychology	Behavioural, oral (amitriptyline, PPS), intravesical, Hunner fulguration, BTX-A; diversion last

E · GU TUBERCULOSIS IN FOUR LINES

ALGORITHM

1. **Suspect:** sterile pyuria, contracted bladder, beaded vas.
2. **Confirm:** three early-morning urines for AFB, culture, GeneXpert.
3. **Treat:** 2HRZE, then 4HR; watch for ureteric stricture.
4. **Operate** after 4–6 weeks of drugs (see Ch 31).